

Final Summary HCI

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YESSSSSSS I am finally done with this summary, and so are you! Well done!!!!!! 🙌

Good luck with the exams!

Notes

Hello hello, welcome to another summary :)

This one contains all the mandatory chapters/paragraphs and important lecture notes.

Good luck!

Chapter 1 - Historical Context

1940 → computers.

1980 → interaction (with average humans) became an issue.

1.1 Introduction

1980s → HCI emerged

Owes a lot to other disciplines, e.g. field of **human factors (ergonomics)**.

Human factors:

- A science and a field of engineering.
- Concerned with:
 - Human capabilities, limitations and performance.
 - Design of systems.

HCI is human factors, but focused on human interaction with computing technology (but still has an interdisciplinary scope).

SIGCHI: interest group on computer-human interaction sponsored by the **ACM**.

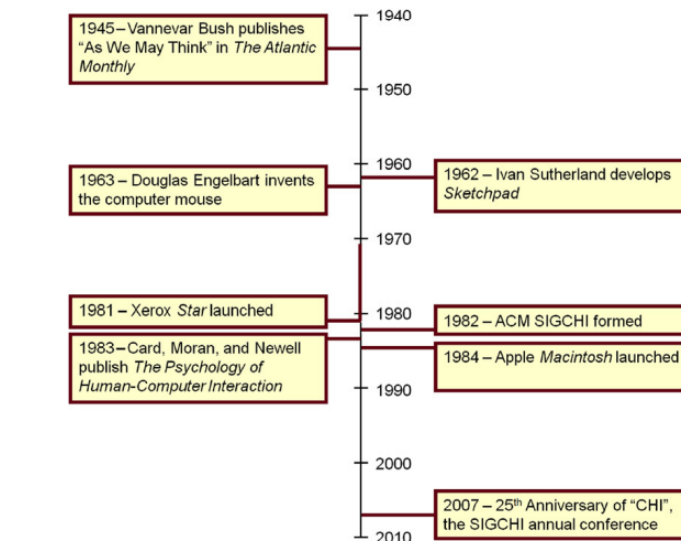


FIGURE 1.1

Timeline of notable events in the history of human-computer interaction HCI.

1.2 Vannevar Bush's "as we may think" (1945)

Essay concerned the dissemination, storage, and access to scholarly knowledge.

Hard to keep up with latest advantages in the information age.

Bush proposed navigating the knowledge maze with a device he called **memex**.

Features of **memex**:

- Associative indexing
 - Points of interest can be connected and joined so that selecting one item immediately and automatically selects another.

1.3 Ivan Sutherland's Sketchpad (1962)

Sketchpad: graphics system that supported the manipulation of geometric shapes and lines on a display using a light pen.

- Commands were not typed.

- Objects were drawn, resized, grabbed and moved, extended, deleted directly using the light pen.
- The first direct manipulation interface.

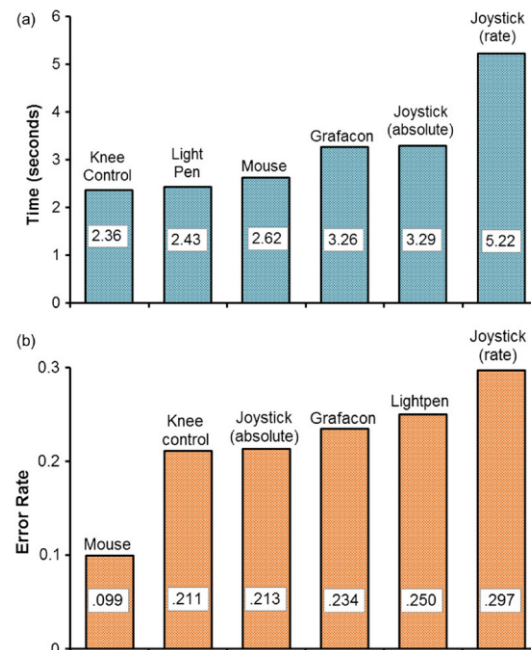
1.4 Invention of the mouse (1963)

Computer mouse symbolizes the emergence of HCI.

- Instead of typing commands, a user could manipulate a mouse to control an on-screen tracking symbol.
- NLS (hypertext system), needed an improved pointing device.
 - The light pen needed to be replaced.
- Close to the keyboard were the users hands are.

Grafacon: a commercial device used for tracing curves.

Results of measured users' access time and motion time (empirical milestone in HCI):



The mouse was the clear winner in terms of accuracy.

1.5 Xerox star (1981)

The first commercially released computer system with a GUI.

- Had windows, icons, menus, and a pointing device (WIMP).
- Supported direct manipulation and what-you-see-is-what-you-get (WYSIWYG) interaction.
- Had what was needed to bring computing to the people.
- The display was bit-mapped (images were formed by mapping bits in memory to pixels on the display).
 - Most systems at the time used character-mapped displays (less memory, limited richness).
- One novel feature: use of the desktop metaphor (brings concepts from the office desktop to the system's display. On the display the user finds icons and accessories for things like documents and calculator). Metaphors are important in HCI.
- Interactions were created that deal with files, not programs, to make the system usable (invisible). Users "open a document," rather than "invoke an editor." Files are associated with applications, but these details are hidden from the user.
- With a GUI and point-select interaction, the Star interface was the archetype of direct manipulation.
 - Before, the user had to learn and remember the syntax of the system's commands and type them in to get things done.
 - **Continuous property:** e.g. display brightness or sound volume, has a continuous control, such as a slider.
 - **Discrete property:** e.g. font size or family, has a discrete control, such as a multi-position switch or a menu item.
 - **Command-line interfaces**, by comparison, are simple. They follow **sequential programming**. Every action occurs in a sequence under the system's control.

- Direct manipulation interfaces require a different approach because they must accept the user's actions according to the user's wishes.
- The Star was not a personal computer, but a networked workstation connected to a server and intended for an office environment.

1.6 Birth of HCI (1983)

1983 → birth of HCI.

- Key events:
 - First **ACM SIGCHI** conference.
 - Publication of Card, Moran, and Newell's **The Psychology of Human-Computer Interaction**.
 - Announcement of the **Apple Macintosh**.

1.6.1 First ACM SIGCHI conference (1983)

1969 → SIGSOC was formed.

- First: focused on computers in the social sciences.
- Later: focused on the needs and behavioral characteristics of the users.
- Statement about SIGCHI and its mission:
 - Association of professionals computer-human interaction.
 - Interdisciplinary group of computer scientists, software engineers, psychologists, interaction designers, graphic designers, sociologists, and anthropologists, etc.

1983 → first SIGCHI conference.

- Slightly modified name to reflect its new stature: ACM SIGCHI Conference on Human Factors in Computing Systems.
- Conference brings together both researchers and practitioners.

1.6.2 The psychology of human-computer interaction (1983)

Psychology of Human-Computer Interaction (book by Stuart Card, Tom Moran, and Allen Newell).

- Emerged from work done at Xerox PARC.
 - PARC's Applied Information-Processing Psychology Project (AIP) mission:

"to create an applied psychology of human-computer interaction by conducting requisite basic research within a context of application".
 - 13 chapters:
 - Scientific foundation (100 pages)
 - Text editing examples (150 pages)
 - Modeling (80 pages)
 - Extensions and generalizations (100 pages)
- Applied psychology: built upon basic research in psychology.
- For many students, the book was their first formalized exposure to human perceptual input, cognition, and motor output.
- **Model human processor (MHP)**: had an eye and an ear (for sensory input to a perceptual processor), a brain (with a cognitive processor, short-term memory, and long-term memory), and an arm, hand, and finger (for motor responses).
- Text editing seemed like the right framework in which to develop an applied psychology of human-computer interaction. All the issues pertinent to text editing are applicable across a broad spectrum of human-computer interaction.
- An interesting synergy between psychology and computer science, is the notion that human behavior can be understood, even modeled, as an information processing activity.
- Adapted information processing models of human behavior to interactive systems.

- Two most prominent examples in the book:
 - Hick's law for choice reaction time.
 - Fitts' law for rapid aimed movement.
- Dilemma in the field: experimentation cannot be done until it is too late.
- An objective in the book was to generate "tools for thought." → casual reference to models of interaction).
 - The book's main contribution on modeling, was to convincingly demonstrate why and how models are important and to teach us how to build them.
- Suggests using approximate "back of the envelope" calculations as a convenient way to describe or predict user interaction.

Physical matching: ensures that the physical attributes of a user align with the design of a computer system or device.

1.6.3 Launch of the Apple Macintosh (1984)

Was sleek and used the latest input device, a computer mouse.

The operating system and applications software introduced the new age of the GUI with direct manipulation and point-select interaction.

The interface was simple (part of this was the one-button mouse) and intuitive. Anyone could use it.

1.7 Growth of HCI and graphical user interfaces (GUIs)

GUIs entered the mainstream and a much broader community of users and researchers were exposed to this new genre of interaction.

1.8 Growth of HCI research

Research initially about the quality, effectiveness, and efficiency of the interface.

Recognition is preferred over recall in user interfaces.

Menu design issues were actively pursued as research topics in the early days of HCI.

Chapter 2 - The Human Factor

The variability of humans means our work is never precise.

The more we understand humans, the better are our chances of designing interactive systems interactions that work as intended.

2.1 Time scale of human action

Time is the most common dependent variable in experimental research in HCI.

Scale (sec)	Time Units	System	World (theory)
10^7	Months		SOCIAL BAND
10^6	Weeks		
10^5	Days		
10^4	Hours	Task	RATIONAL BAND
10^3	10 min	Task	
10^2	Minutes	Task	
10^1	10 sec	Unit task	COGNITIVE BAND
10^0	1 sec	Operations	
10^{-1}	100 ms	Deliberate act	
10^{-2}	10 ms	Neural circuit	BIOLOGICAL BAND
10^{-3}	1 ms	Neuron	
10^{-4}	100 μ s	Organelle	

FIGURE 2.1

Newell's time scale of human action.

The higher in the scale, the more qualitative research. The lower, the more quantitative research.

2.2 Human factors

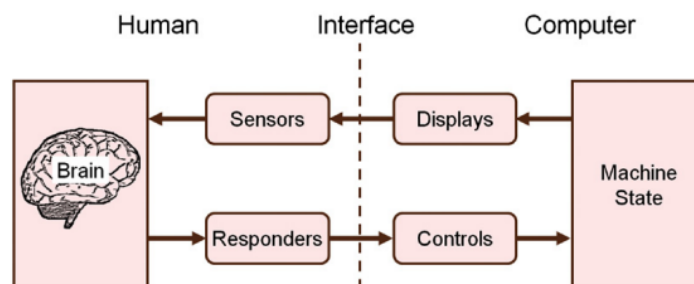


FIGURE 2.2

Human factors view of the human operator in a work environment.

Model showing a human operator confronting a machine.

The human monitors the state of the computer through sensors and displays and controls the state of the computer through responders and controls.

Dashed vertical line is the interface where interaction takes place.

- This is the location where researchers observe and measure the behavioral events that form the interaction.

2.3 Sensors

Five classical human senses: vision, hearing, taste, smell, and touch.

Reception and conversion into electrical nerve signals of physical phenomena.
The signals are transmitted to the brain for processing.

Sensory stimuli and sense organs are purely physiological.

2.3.1 Vision (Sight)

Vision, or sight: the human ability to receive information from the environment in the form of visible light perceived by the eye.

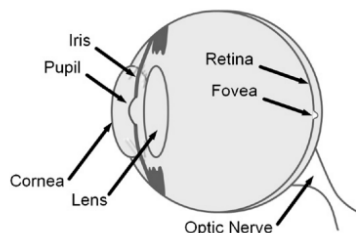


FIGURE 2.3

The eye.

Most people obtain about 80 percent of their information through the sense of light.

The act of seeing:

1. The reception of light through the eye's lens.
2. The lens focuses the light into an image projected on to the retina at the back of the eye.
3. The retina (a transducer) converts visible light into neurological signals sent to the brain via the optic nerve.

4. The fovea is responsible for sharp central vision.

Properties of light:

- **Frequency:** leads to the perception of color.

Visible light is a small band in the electromagnetic spectrum, which ranges from radio waves to x-rays and gamma rays. Different colors are positioned within the visible spectrum of electromagnetic waves, with violet at one end (390 nanometers) and red at the other (750nm).

- **Intensity:** related to intensity is **luminance**, which refers to the amount of light passing through a given area.

With luminance comes **brightness**, a subjective property of the eye that includes perception by the brain. The unit for luminance is candela per square meter (cd/m²).

- **Fixations:** the eyes are stationary, taking in visual detail from the environment. Typically last at least 200ms.
- **Saccade:** rapid repositioning of the eyes to a new position. Takes only 30–120ms.
 - **Fixations** and **saccades:** eye movements.

2.3.2 Hearing (Audition)

Hearing (or audition): the detection of sound by humans. Sound is transmitted through the environment as sound waves (created when physical objects are moved or vibrated, thus creating fluctuations in air pressure).

Human speaking: the physical object creating the sound is the **larynx**, or vocal cords, in the throat.

Hearing occurs when sound waves reach a human's ear and stimulate the ear drum to create nerve impulses that are sent to the brain. Sound has at least four physical properties: **intensity** (loudness), **frequency** (pitch), **timbre**, and **envelope**.

- **Loudness:** the subjective analog to the physical property of intensity. It is quantified by sound pressure level, which expresses the pressure in a sound wave relative to the average pressure in the medium.

- The unit of sound pressure level is the decibel (dB).
 - Hearing begins: **0–10 dB**.
 - Conversational speech: **50–70 dB**.
 - Pain: **120–140dB**.
- **Pitch**: the subjective analog of frequency, which is the reciprocal of the time between peaks in a sound wave's pressure pattern.
 - The units of pitch are cycles per second, or **Hertz (Hz)**.
 - Humans can perceive sounds in the frequency range of about **20Hz–20,000Hz**, the upper limit tends to decrease with age.
- **Timbre** (aka richness or brightness): results from the harmonic structure of sounds. It is the timbre that distinguishes the tone as being from a particular musical instrument.
- **Envelope**: the way a note and its harmonics build up and transition in time (from silent to audible to silent).

Other (non physical) properties of sound that have to do with perception:

- Harmonious (pleasant)
- Discordant (unpleasant)
- Sense of urgency or speed
- Sound's location (direction and distance)

2.3.3 Touch (Tactition)

Somatosensory system: includes sensory receptors in the skin, muscles, bones, joints, and organs that provide information on a variety of physical or environmental phenomena (touch, temperature, pain, and body and limb position).

Tactile feedback (in HCI): information provided through the somatosensory system from a body part, when it is in contact with (touching) a physical object.

- All user interfaces that involve physical contact with a user's body part include tactile feedback.

2.3.4 Smell and taste

Smell (olfaction): the ability to perceive odors.

- Occurs through sensory cells in the nasal cavity.

Taste (gustation): a direct chemical reception of sweet, salty, bitter, and sour sensations through taste buds in the tongue and oral cavity.

Flavor: a perceptual process in the brain that occurs through a partnering of the smell and taste senses.

These senses are not generally “designed in” to systems.

2.3.5 Other senses

The word sense appears in many contexts, e.g. sense of urgency, sense of direction, musical sense, intuitive sense, moral sense, common sense.

These additional senses encapsulate how humans feel about their interactions with computers.

2.4 Responders

Through movement, or motor control, humans are empowered to affect the environment around them. Control occurs through responders.

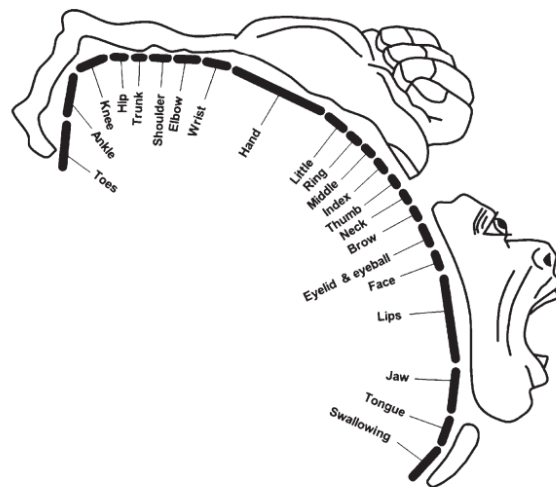


FIGURE 2.9

Motor homunculus showing human responders and the corresponding cortical area.

Maps areas in the cerebral motor cortex to human responders. The lengths of the underlying solid bars show the relative amount of cortical area devoted to each muscle group.

Card et al:

“those groups of muscles having a large area devoted to them are heuristically promising places to connect with input device transducers if we desire high performance,”

“the determinants of muscle performance are more complex than just simple cortical area”

2.4.1 Limbs

Human control over machines is usually associated with the limbs, particularly the upper body limbs. The same is true in HCI.

Limbs can also act as responders and provide input to a computer.

Movement of the limbs is tightly coupled to the somatosensory system.

Proprioception: the coordination of limb movement and position through the perception of stimuli within muscles and tendons.

2.4.2 Voice

Vocal cords are responders. Through the combination of movement in the larynx, or voice box, and pulmonary pressure in the lungs, humans can create a great variety of sounds.

Computer input not only possible through speech, but also possible using non-speech vocalized sounds (non-verbal voice interaction (NVVI)). In this case, various acoustic parameters of the sound signal, such as pitch, volume, or timbre, are measured over time and the data stream is interpreted as an input channel.

2.4.3 Eyes

The human eye receives sensory stimuli in the form of light from the environment.

The eye is also capable of acting as a responder (controlling a computer through fixations and saccades). Then the eye acts both as a sensor and as a responder.

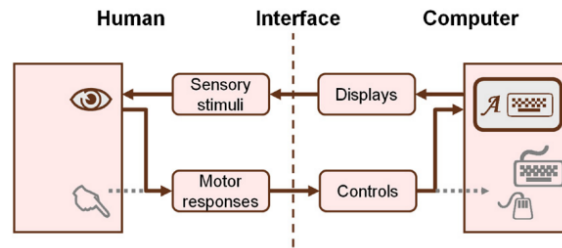


FIGURE 2.12

The human-computer interface with an eye tracker. The eye serves double duty, processing sensory stimuli from computer displays and providing motor responses to control the system.

2.5 The brain

The brain is the most complex biological structure known. While sensors (human inputs) and responders (human outputs) are nicely mirrored, it is the brain that connects them.

2.5.1 Perception

Perception: first stage of processing in the brain, occurs when sensory signals are received as input from the environment.

- At this stage associations and meanings take shape.
 - With associations and meaning attached to sensory input, humans are vastly superior to the machines they interact with.

Psychophysics (specialized area of experimental psychology): examines the relationship between human perception and physical phenomena.

- A common experimental goal is to measure the just noticeable difference (JND) in a stimulus.

Another property of perception is **ambiguity**: the human ability to develop multiple interpretations of a sensory input.

- **Illusion**: the deception of common sense (can be in various senses).

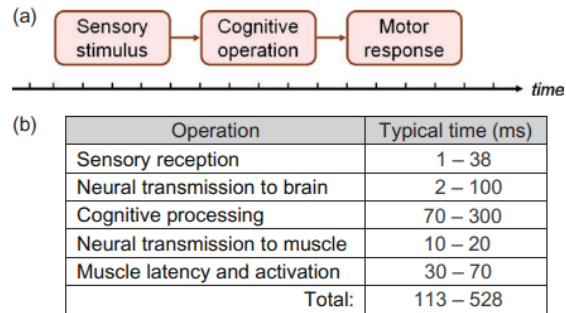


FIGURE 2.17

Cognitive operation in a reaction time task: (a) Problem schematic. (b) Sequence of operations (Bailey, 1996, p. 41).

2.5.2 Cognition

Cognition: the human process of conscious intellectual activity, such as thinking, reasoning, and deciding.

Occurs within the human brain, so studying cognition presents special challenges.

- Beginning and ending of cognitive operations often is not precisely known.
- Difficulty in pin pointing where and how the measurements are made.

2.5.3 Memory

Memory: the human ability to store, retain, and recall information.

There are similarities between memory in the brain and memory in a computer.

- Computer memory often includes separate areas for data and code.
- In the brain, memory is similarly organized.
 - **Declarative/explicit area** stores information about events in time and objects in the external world.
 - Similar to a data space.
 - **Implicit/procedural area** in the brain's memory stores information about how to use objects or how to do things.
 - Similar to a code space.

Within long-term memory is an active area for short-term memory or working memory. The contents of working memory are active and readily available for access (about seven units).

Humans are typically able to distinguish about seven levels of a uni-dimensional stimulus.

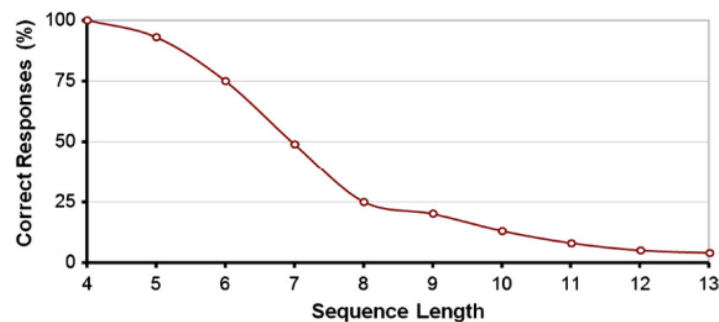


FIGURE 2.18

Results of a test of short-term memory.

Chunking: a process whereby humans group a series of low-level items into a single high-level item (intuitive process).

2.6 Language

Language: the mental faculty that allows humans to communicate.

Language as speech is available without effort. Learning to write does demands effort.

“Humankind is defined by language; but civilization is defined by writing.”

In HCI, our interest in language is primarily in systems of writing and in the technology that enables communication in a written form.

Text: the written material on a page or display.

Corpus: a large collection of text samples gathered from diverse and representative sources.

- Not possible for a corpus to broadly yet precisely represent a language. The sampling process brings limitations.
- A corpus is sometimes reduced to a word-frequency list.

- A notable feature of some corpora is part-of-speech (POS) tagging, where words are tagged by their category.
 - The part of speech is contextual, reflecting a word's use in the original text.

2.6.1 Redundancy in language

- We automatically insert words that are omitted or obscured (ham and ____ sandwich).
- We anticipate words (a picture is worth a thousand ____), or entire phrases (to be or ____ _).

2.6.2 Entropy in language

If redundancy in language is what we inherently know, **entropy** is what we don't know: the uncertainty about forthcoming letters, words, phrases, ideas, concepts, etc.

Redundancy and entropy are related: If we remove what we know, what remains is what we don't know.

Errors are more common at the beginning of words, less common as words progress.

Additional notes lecture week 1

Human-Computer Interaction

"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them

Human factors

Vision/sight in many interaction scenarios the (most) important sense.

- Often not passive: eyes move in fast series of fixations and saccades.

- Eye tracking often used in HCI studies.

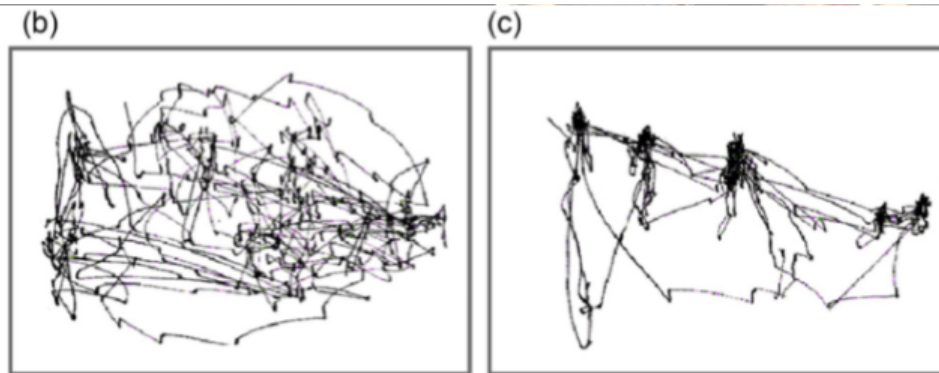


FIGURE 2.6

Yarbus' research on eye movements and vision (Tatler et al., 2010). (a) Scene. (b) Task: *Remember the position of the people and objects in the room.* (c) Task: *Estimate the ages of the people.*

Hearing/audition increasingly important

- Devices increasingly used for listening to music.
- Speech recognition and speech generation.
- Accessibility when visual interfaces are not suited.

Touch/tactition

- (Passive) identifiers on keyboards (F/J).
- Resistance keys, mouse buttons, joysticks.
- (Active) vibration phones, game controllers.

Responders

- Limbs
- Voice
- Eyes
 - Eye typing: eye as sensor and responder

Human performance

- Reaction time

- Visual search
- Practice & skill
- Attention
- Human Error

Chapter 3 - Interaction Elements

If the user is engaged in any activity with computing technology, **interaction** is taking place.

- The human systems at work are **deliberate acts** ($\approx 100\text{ms}$), **operations** ($\approx 1\text{ s}$), and **unit tasks** ($\approx 10\text{ s}$).
- These tasks are well suited to empirical research due to their short duration.
 - Experiments with high internal validity are possible.

Interaction: humans using their sensors and responders to monitor and control devices, machines, and systems that include computing technology.

- The internal states of the human and the machine interact in a closed-loop system through **controls and displays** (the machine side) and **motorsensory behaviors** (the human side).
 - Inputs are inputs to the machine that are controlled or manipulated by human outputs (e.g. limbs and speech).

3.1 Hard controls and soft controls

Through soft interfaces, the way humans interact with technology changed → the **malleability** (the quality of something that can be shaped into something else without breaking) of a display (through software) brings unlimited possibilities to a relatively small physical space. The result is soft controls.

Human-computer interfaces contain lots of controls. Most are soft controls manipulated by physical controls (the mouse or keyboard).

- Controls invoke different responses (displays) depending on the context. The distinction between controls and displays is blurred with soft controls.

- Soft controls their size, shape or appearance can change dynamically to convey information to the user.

Each of these soft controls is manipulated by a hard control, by clicking, double-clicking, etc.

Space multiplexing: there is a dedicated physical control for each parameter that is controlled. Each control has a separate physical location.

Time multiplexing: there are fewer controls than parameters, and so a single device is reconfigured to control different parameters at different stages of an operation.

3.2 Control-display relationships

In HCI the relationship between what a user does and what is experienced is supposed to be natural, seamless, intuitive, and efficient.

Control-display relationships are some times called **mappings**.

Spatial relationship: the strategic arrangement and positioning of interface elements to enhance usability, navigation, and user experience.

Dynamic relationships: interactive and responsive connection between a user and the interface.

Physical relationships: interactions between users and input/output devices, influencing usability, accessibility, and ergonomic design considerations.

3.2.1 Spatial relationships

The mouse-cursor spatial relationship:

- The mapping is **congruent** because there is an exact spatial correspondence between the controller input and the display output. Move the mouse right, and the cursor moves right. Move the mouse "forward" and the cursor moves "up."

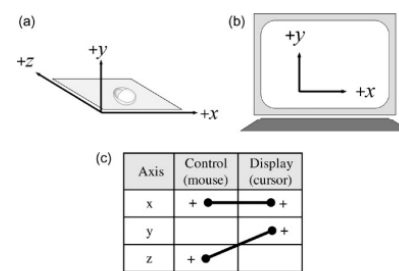
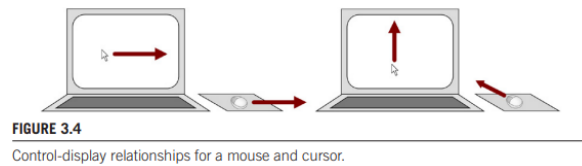


FIGURE 3.5
Axis labels. (a) Control space. (b) Display space. (c) Control-display mapping for a mouse and cursor.

- The plus (+) symbols indicates that positive motion of the control yields positive motion in the display along each axis.
- The y-axis cursor motion is an example of a **transformed spatial mapping**. In this case the transformation is 90°, if viewed along the y-z plane. Aiming error is higher for 90° to 135° transformations and lower for 0° or 180° transformations. Adaptation occurs for all transformations, in a few as 50 trials.

Scrolling the view of an image or document:

- A scroll pane includes scrollbars with sliders to move the view. Interaction involves manipulating a **physical controller**, such as a mouse (**hard control**), to move a pointer to the slider (**soft control**) and acquiring it with a button-down action.

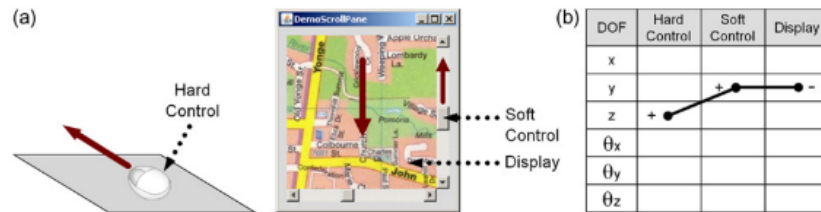


FIGURE 3.6

A three-tier control-display relationship: (a) Moving the hard control forward moves the soft control up, which in turn moves the display view down. (b) Control-display mappings.

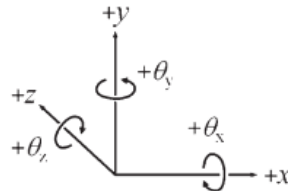


FIGURE 3.7

Axis labels for a three-dimensional space.

- The **theta (θ)** is used to designate angle or rotation. The arrows indicate positive rotation, corresponding to clockwise movement as viewed looking out from the origin along each axis.
- There are six parameters, or **degrees of freedom (DOF)**.
 - **Degrees of freedom:** each parameter may be manipulated independently of the others.
 - To fully place an object in 3D space, six parameters are required: three for the object's position in space (x, y, z), and three for the object's orientation in space (θ_x , θ_y , θ_z).

Example where **congruence** is fully attained:

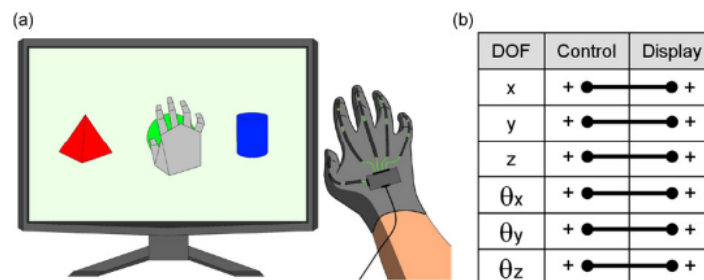


FIGURE 3.8

Spatial congruence in 3D: (a) 6 DOF input controller and 6 DOF display (sketch courtesy of Bartosz Bajer). (b) 6 DOF control-display correspondence.

Rotation is a challenge since traditional controllers only sense translation along the x-axis and z-axis. Control-display spatial congruence is not possible.

- Typical solution is modes (commonly panning (rotating camera left-right and up-down), zooming, and positioning).
 - Another way to pan with the mouse is through the soft control labeled Pan in the figure, however, it requires a tight coupling of left-right, up-down linear movement of the mouse and the user's visual attention (on the soft control).
- Using y-axis rotation is a potentially interesting and new interaction technique.



FIGURE 3.9

Rotating a 3D scene: (a) Street scene in Toronto with pan and zoom controls. (b) Mapping of mouse control for panning if the scene is dragged.

3.2.2 CD gain and transfer function

Property to mouse-cursor movement:

- **CD gain:** the amount of movement in a display object, such as a cursor, for a given amount of movement in a control.
 - **CD gain = amount the cursor moved/amount the mouse moved**
 - Often the relationship between controller motion and cursor motion is non-linear and uses a power function → the amount of cursor motion depends on the **velocity** of mouse motion as well as controller motion.
 - This can be changed by the user and if "Enhance pointer precision." is selected, if the mouse moves quickly, CD gain increases. If the mouse moves slowly, CD gain decreases.
 - The term **transfer function** is sometimes used since non-linear relationships are more elaborate. To ensure the cursor is responsive to mouse movement, with no perceivable delay, the software implementation

of a non-linear relationship typically uses a **lookup table** to map controller movement to display movement.

Varying CD gain → trade-off between **gross positioning time** (getting to the vicinity of a target) and **fine positioning time** (final acquisition). The optimal setting is that which minimizes the combined gross and fine positioning times.

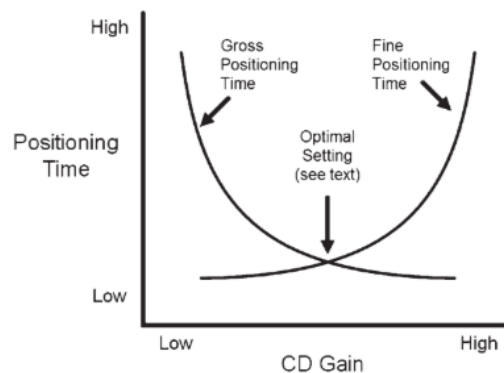


FIGURE 3.11

Trade-off between gross positioning time and fine positioning time as a function of CD gain.

Other factors such as display size or scale (independent of the CD gain setting) also bring into question the optimization of this common input device parameter.

Studies found that significant performance benefits are not achieved by adjusting CD gain.

- What is optimal performance (what is the goal)? The goal of optimizing both speed and accuracy is problematic, because of the speed-accuracy trade-off.

3.2.3 Latency

Subsequent input depends on the latest output. Human performance and the interaction experience are adversely affected when feedback is delayed.

Latency (or lag): the delay between an input action and the corresponding response on a display.

- Often present with Internet connections. The research issues here are more about user experience and frustration than about human performance. E.g. frustration is reduced if the interface includes feedback informing the user that the network is processing the request.

- Latency is attributable to properties of input devices, output devices, and software. Latency increases further due to software overhead (a variety of system-related factors).
- Latency will increase dramatically when output requires substantial computation for graphics rendering.
 - Since the construction of the frame begins only once the position is known, the potential for parallel processing is limited.

It is not possible to accurately measure latency from within the target system. Using software within a system misses the actions and responses at the human interface. To reflect the user's experience as a whole, a proper latency measurement is done externally. Example:

- The input controller swings in front of the output display with a high-speed camera recording the temporal relationship between input motion and the output response. Latency in virtual reality is associated mostly with 6 DOF tracking devices attached to the hand, head, or body as well as the latency caused by the low frame rates.



FIGURE 3.12

Polhemus G⁴™ 6 DOF tracking system. The 6 DOF sensor is identified by the arrow.

These devices and their interfaces transmit six degree-of-freedom position and orientation data to the host computer while acting in concert with competing processes in the complete VR environment. Maintaining negligible latency is usually not possible.

Although latency is known as a compromising parameter in VR systems, and its effect on human performance is considerable, the evidence for this is mostly

anecdotal. Low latency is significantly more important than high-quality graphics or stereoscope.

- Screen updates are needed 15 times per second minimum. Even at this rate, subjects noted the "sponginess of the feel".

Empirical measurements of the effect of latency on human performance are rare.

- There was some research in the 1960s on the design of controllers for remote manipulation systems; however, the latency tested was extremely long.

Since latency is on the order of a few hundred milliseconds in VR, its effect on human performance is less apparent.

Study:

With 75ms latency, movement time increased by 16 percent and error rate by 36 percent. At 225ms the effect was dramatic, with movement time increasing by 64 percent and error rate by 214 percent.

3.2.4 Property sensed and order of control

When a human engages an input controller, the input controller senses the interaction and converts a property sensed into data that are transmitted to the host computer for processing.

Most common properties sensed for pointing devices:

- Position
- Displacement
- Force

With a graphics tablet, touchpad, or touchscreen, the property sensed is the **position** of a stylus or a finger on a digitizing surface.

- Position is sensed as an absolute coordinate at the point of contact along the x-axis and z-axis.

With a mouse, the property sensed is the **displacement** (the amount of movement along the x-axis and z-axis).

Touchpads in notebook computers typically operate in **mouse-emulation mode**. Even though the property sensed is the absolute position of the finger on the

touchpad surface, the value is reported to the host in relative terms (the amount of finger displacement relative to the last sample).

Most common orders of control:

- **Position-control** (zero-order control): the sensed property of the input device controls the position of the object or view on a display (e.g. a mouse).
- **Velocity-control** (first-order control): the sense property controls the velocity of the object or view.

Order of control is often associated with joysticks. Joysticks are either **isotonic** or **isometric**.

- **Isotonic joystick (or displacement joysticks)**: the user manipulates the stick handle, which, in turn, swivels about a pivot-point. The property sensed is the movement of the stick.
- **Isometric joystick**: the stick does not move. The property sensed is the force applied to the stick.

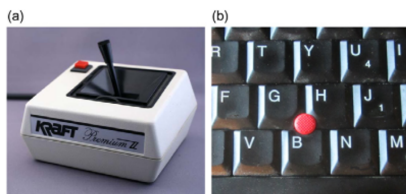


FIGURE 3.13
Joysticks: (a) Isotonic (displacement sensing).¹¹ (b) Isometric (force sensing).

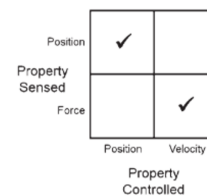


FIGURE 3.14
Order of control mapping for property sensed versus property controlled. Checks indicate the preferred mappings.

Research about isometric joystick in both position-control and velocity-control modes showed that velocity-control performed best.

Position-sensing and force-sensing input controllers must be tested in both position-control and velocity-control modes. Position control is best for a position-sensing device and velocity control is best for a force-sensing device.

3.3 Natural versus learned relationships

Spatial transformation between the control motion and the display response → need to learn a relationship.

- If the relationship is expected and considered correct by a majority of people within a population, then the relationship holds (accepted and considered natural by that population (still a learned relationship)). It is more accurately a population stereotype, or cultural standard.

“Warrick’s principle”: the display indicator should move in the same direction as the nearest point on the control (example of applied ergonomics, little basis in human sensory-perceptual cognition).

Wherever there is a spatial transformation, the ensuing relationship is learned, not natural.

The best possible scenario is **spatial congruence** (i.e. upward movement in the control handle produces upward movement in the display reading).

A tenable physical analogy is difficult to form in some situations. Given a consistent recurring implementation within a geographical region, a population stereotype will emerge.

If there is a physical contradiction, people would learn the relationship, because they must. But they would make more errors than if the relationship was based on a correct physical mapping.

3.4 Mental models and metaphor

Learning and adaptation often through physical analogy or metaphor.

- It is important to find a spatially congruent physical understanding. Physical analogies and metaphors are examples of the more general concept of mental models, also known as **conceptual models**. An association with human experience is required.

HCI’s first mental model: desktop metaphor, helped users understand the graphical user interface.

Implementation models are to be avoided. These are systems that impose on the user a set of interactions that follow the inner workings of an application.

It is desirable to exploit pre-existing experience at every opportunity in designing a software-based product. Field popping up while hovering over an icon providing a terse elaboration on the button’s purpose, can help for a better experience (tooltips or screentips).

3.5 Modes

Mode: a functioning arrangement or condition.

- Usually, modes are viewed in a more limited sense, such as lowercase mode versus uppercase mode.
- For interactive systems, challenges with modes occur because there are fewer controls than tasks (e.g. key board not a lot of keys but lots of key variations).

Function keys have many modes.

Physical keys on standard keyboards are hard keys. They have a few modes.

Soft-hard keys: involve both a physical key and a definition of the key created through software. They have lots of modes.

Soft keys have an infinite number of modes.

Designing modeless interactions is sometimes viewed as desirable. But modes can't be avoided in most cases.

From a design perspective, the main issues with modes are:

- a. Changing modes
 - Clicking the desired soft button. It's all quite simple, and it works reasonably well. But still problems lurk.
 - With more modes, come more transitions. Not all users know these transitions.
- b. Feedback, or making the mode apparent to the user.
 - The issue is the quality of feedback to the user (informative vs visible feedback).
 - Problems arise if the user doesn't know what mode or state the system is in.

Modes are common in graphics and 3D interaction.

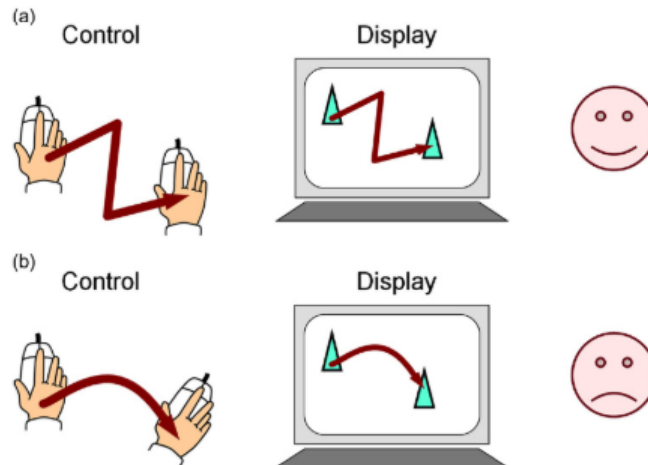


FIGURE 3.26

Two-dimensional movement with the hand + mouse: (a) x and y displacement only. (b) z -axis rotation of the hand + mouse is not sensed, so the object on the display undergoes x and y displacement only.

The typical fix is a rotate mode or to re-engineer a mouse to sense rotation.

Design issues for 3 DOF mice:

- usually the extra DOF is not needed. When a task requires both translation and rotation, a modifier key can engage the third DOF. This analysis began as an effort to eliminate the rotate mode, yet the solution includes modes.
- Accommodating the biomechanics of the lower arm. It is difficult to effect the required wrist movement if an on-screen object is rotated through a wide angle. A solution is to use another modifier key to amplify the angular mapping of mouse rotation. This is somewhat analogous to CD gain, except the gain is applied to rotation. Another approach is a camera-based system, with a mouse pad serving as the ground plane for the camera image. Two-handed technique: one hand manipulates and rotates the mouse, while the other hand counter-rotates the mouse pad.

The situation is more complicated in 3D environments, since there are six degrees of freedom. If the input controller has fewer than six degrees of freedom, then modes are needed to fully access the environment.

If 3D is the goal, then a more sophisticated setup is needed. Typically, different or additional devices are used to provide more natural control-display relationships.

3.6 More about degrees of freedom

Examples of devices engineered to provide additional sensing capabilities:

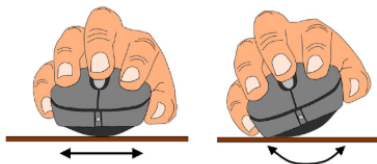


FIGURE 3.31

The Rockin' Mouse works like a mouse but also senses side-to-side (θ_x) rotation (shown) and forward-backward (θ_y) rotation. (Balakrishnan et al., 1997). Rotation about the y -axis is not sensed.

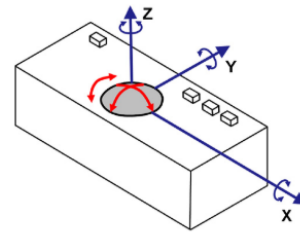


FIGURE 3.32

Three-axis trackball.

It is important to consider the relationship between device properties, such as degrees of freedom, and task requirements.

Integrity and separability of the degrees of freedom: with a regular mouse, controlling one degree of freedom in isolation of the other is difficult.



FIGURE 3.33

Etch-A-Sketch children's toy with separate controls for the x and y degrees of freedom.¹⁸

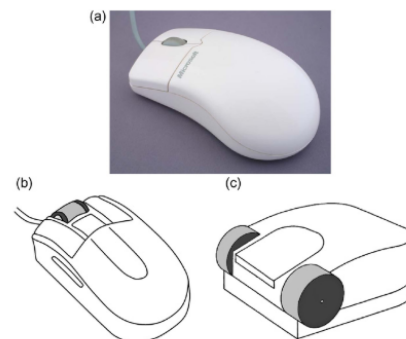


FIGURE 3.34

Wheel mice: (a) Microsoft's Intellimouse. (b) Mouse Systems' ProAgio. (c) Roller Mouse research prototype.

(Source: b, adapted from Gillick and Lam, 1996; c, adapted from Venolia and Ishikawa, 1994)

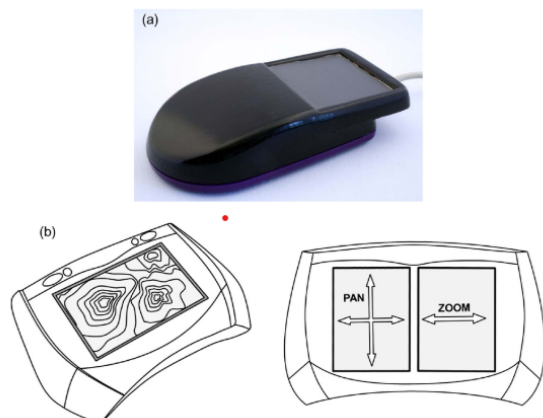


FIGURE 3.35

Adding degrees of freedom with a touchpad: (a) PadMouse (Balakrishnan and Patel, 1998).¹⁹ (b) Zooming and panning display screen (Silfverberg, Korhonen, and MacKenzie, 2003), top view (*left*), bottom view (*right*).



FIGURE 3.36

Multi-touch sensing input devices (Villar et al., 2009).

Degrees of freedom (DOF) is a property that spans a rich complement of tasks and interaction techniques. User interactions are plentiful, and they are juxtaposed in complex ways, such as overlapping, sequential, and in parallel.

3.7 Mobile context

Arrival of Apple iPhone → landscape of HCI experienced a dramatic shift. Pressing keys on small keyboards or gesturing with a stylus gave way to finger actions: swiping, flicking, pinching, tapping, and so on. Devices in this genre include additional components to enhance the user experience.

Many of the interaction elements for desktop environments don't exist in the mobile context. This follows from the distinction between indirect input (the system response is spatially displaced from the input device) and direct input (the system response is located at the point of touch, cursor is not needed → no control-display mappings or CD gain).

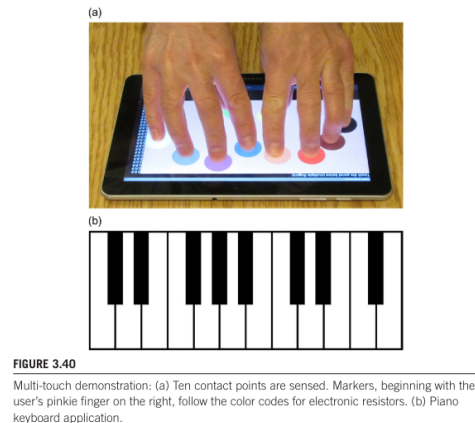
Direct touch input is the full embodiment of WYSIWYG interaction (what you see is what you get).

Problem: with touchscreens, users typically do not know, and cannot control, the precise location of the contact point. Two reasons:

1. The finger occludes the location of contact
2. Finger contact forms an area, not a point.

- These issues are sometimes called the **“fat finger problem”**.

Another difference between touch input and mouse input is the ability of touch-sensing technologies to sense multiple contact points. Recent devices can sense several contact points.



For the piano, with added pressure data, both the pitch and loudness of notes can be controlled upon finger contact.

Today's mobile devices include a host of additional components:

- An accelerometer (allows the angle or tilt of the device to be sensed).
 - As with any new technology, research initially focused on what can be done rather than on how well it can be done.

3.8 Interaction errors

Empirical research in HCI is largely about finding the physical properties and combinations that improve and enhance human performance.

- Although the time to complete a task can enhance or hinder by degree, errors only hinder. Absence of errors is, for the most part, invisible.
 - Interaction errors, are germane to HCI.

Difficult problems (in desktop computing) are solved, and now researchers should focus on new frontiers. This view is partially correct. But desktop computing still has a lot of problems.

First example of interaction error:

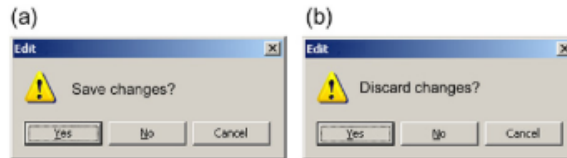


FIGURE 3.42

HCI has come a long way: (a) Today's UIs consistently use the same, predictable dialog to alert the user to a potential loss of information. (b) Legacy dialog rarely (if ever) seen today.

Second example:

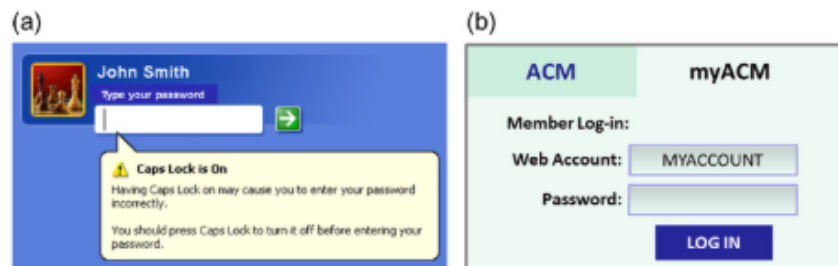


FIGURE 3.43

Entering a password: (a) Many systems alert the user if CAPS_LOCK is on. (b) Others do not.

Another problem:

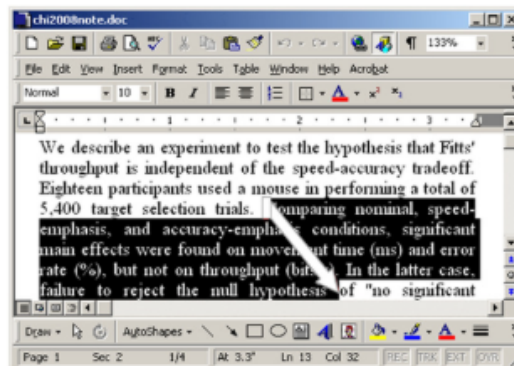


FIGURE 3.44

On the brink of hyper-speed scrolling. As the mouse pointer is dragged toward the edge of the viewable region, the user is precipitously close to losing control over the speed of dragging.

This is a design flaw.

For buttons, focus is usually indicated with a dashed border. For input fields, it is usually indicated with a flashing insertion bar ("|").

- **“Focus advancement”**: the progression of focus from one UI component to the next.
 - There is wide-spread inconsistency in current applications in the way UI widgets acquire and lose focus and in the way focus advances from one component to the next.

A problem that can arise: the user’s attention is on the keyboard while the physical presentation is on the system’s display.

- Absence of expectations keeps the user on guard. The result is a slight increase in the attention demanded during interaction, which produces a slight decrease in transparency. Instead of engaging in the task, attention is diverted to the needs of the computer.

Where the consequences of errors are small, errors tend to linger. The little errors persist.

“Microstrategies focus on what designers would regard as the mundane aspects of interface design; the ways in which subtle features of interactive technology influence the ways in which users perform tasks”.

- User experiences exist as collections of microstrategies. User actions form the experience. It is unfortunate that they often exist simply to serve the needs of the computer or application.

Another reason little errors tend to linger is that they are often deemed user errors, not design, programming, or system errors (**design-induced errors**).

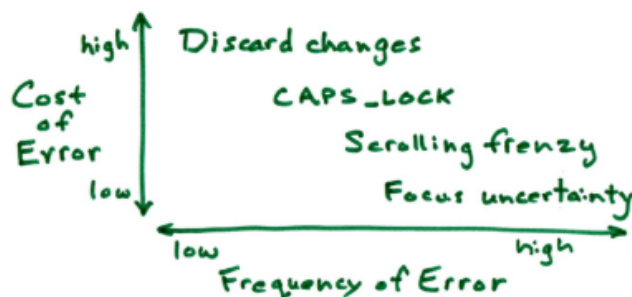


FIGURE 3.46

Trade-off between the cost of errors and the frequency of errors.

High-cost errors occur with low frequency receive a lot of attention and they get dealt with. As systems mature and the big errors get fixed, designers shift their efforts to fixing less costly errors. Over time, more and more systems include reasonable and appropriate implementations of these interactions. Divergence in the implementations diminishes and, taken as a whole, there is an industry-wide coalescing toward the same consistent implementation. The ground is set for user expectation to take hold.

In many ways, the little errors are the most interesting, because they slip past designers and programmers.

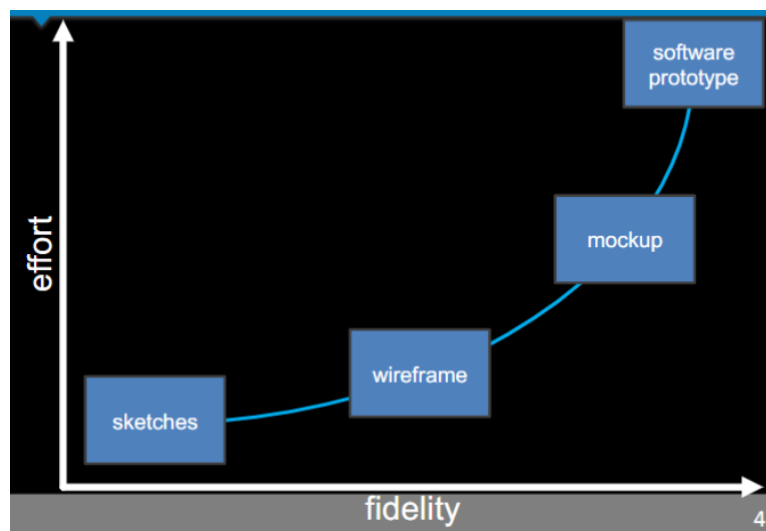
Additional notes lecture week 2

Prototypes

Prototype: A concrete representation of a part or all of an interactive system.

Done to:

- Minimize the time for early design iterations
- Make errors quickly!
- Enables to explore more design options

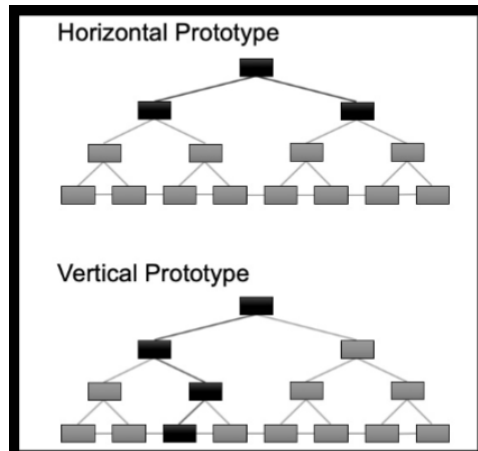


Horizontal prototyping:

- Demonstrates the feature spectrum
 - without full implementation of all individual features
- Used to evaluate
 - Navigation (e.g. finding a specific function or feature)
 - Overall user interface concept
 - Feature placement
 - Accessibility
- User preferences
- Applicable in low-fidelity prototyping and high-fidelity prototyping
- Used in early design stages
 - To determine the set of features to include
 - To decide on the user interface concept

Vertical prototyping

- Demonstrate a specific feature
- The details of the function/feature are shown/implemented
- Used to evaluate/test
 - The optimal design for a particular function
 - Optimize the usability of this function
 - User performance for this particular function
- Often used in high-fi prototyping but is also applicable to low-fi prototyping



Low-fidelity prototypes

- Check ideas and interaction flow
- Fast, cheap and easy to change
- Advantages
 - Cheap, easy and quick to implement
 - Users are keen to criticise
- Disadvantages
 - No real functionality, difficult to identify errors
 - Reuse and extending difficult to impossible
 - Not all ideas can be realized
- Paper prototypes
 - Advantages:
 - Are cheap and easy to do
 - Allow very quick iterations
 - Help to find general problems and difficult issues
 - Provide means for early “testing” with real users
 - Enable non-technical people to interact easily with the design team (no technology barrier for suggestions)

- Make mistakes on paper and make them before you design your architecture and before you do the coding
- Helps to structure the information architecture and hence should lead to better system design
- Difficulties:
 - Feedback depends strongly on how the prototype is explained and handled
 - Only specific use cases can be tested
 - For technical people it is often hard to believe that it works (needs convincing to engage with paper prototypes)
 - Limits what can be prototyped (architecture, physical prototyping, 3D)
 - Limits what can be tested (e.g. the haptic feedback of a new gesture interaction can not be tested)

High-fidelity prototypes

- Looks & feels like the final product
 - Colors, screen layout, fonts, ... text used
 - Response time and interactive behavior
- Restricted functionality
 - Only certain functions work
 - Functionality is targeted towards the tasks
 - Invisible issues (e.g. security)
- Standard technologies for prototyping
 - HTML, JavaScript
 - Axure, Director, Presentation programs
 - GUI Builder (e.g. Visual Basic, Delphi, NetBeans)
- Advantages:
 - Show how the finished product will look and feel

- Behaves like the final product for selected tasks
- Allows predicting efficiency
- Disadvantages:
 - Can be very time-consuming to implement
 - Users can only “use” the implemented functions
 - Feedback centered around look & feel
 - Managers may expect that “the product” is nearly ready

To encourage feedback that is not related to the look & feel it may be helpful to make the GUI look rough.

Physical computing prototypes

- Prototyping physical interaction with Arduino / RaspberryPi
 - Support a huge variety of sensors / actors
 - Easy to use
- Prototyping product appearance
 - 3D print
 - Laser cut
- Follows the trend in HCI towards interactive and personalized products
 - Novel fabrication possibilities, novel materials
 - Fab labs (Waag Society Amsterdam)
 - Huge online community (<http://www.arduino.cc/>
<http://www.instructables.com/>)

Interaction elements

Input: an input device is a transducer from the physical properties of the world into the logical parameters of an application.

Basic Input Operations and the design space:

- Text Input
- Direct Mapped Controls

- Continuous
 - Keyboard Handwriting
 - Spoken
- Block
 - Scan/digital camera and OCR
- Pointing & Selection
 - Degree of Freedom (1, 2, 3, 6, DOF)
 - Isotonic vs. Isometric
 - Transfer function
 - Precision
 - Feedback
- Hard wired buttons/controls
- On/off switch, Volume slider
- Physical controls that can be mapped
- Function key on keyboard
- Industrial applications
- Media and Sensor Capture
 - Media type
 - Audio
 - Images
 - Video
 - Sensor type
 - Quality/Resolution

Classification of Pointing devices

- Dimensions
 - 1D / 2D / 3D
- Direct vs. indirect
 - integration with the visual representation
 - Touch screen is direct
 - Mouse is indirect
- Discreet vs. continuous
 - one single unit vs. a sequence of information (in x DOF) interpreted as one input action
 - Push button is discreet
 - Mouse is continuous (mouse's position, though digitized into a discrete quantity, is fast enough to be considered continuous)

- Absolute vs. Relative
 - movement/position used as input
 - Touch screen is absolute
 - Mouse is relative

Controller Resistance (2DOF, 6DOF)

- Isometric
 - Device/handle is not moved
 - pressure devices / force devices
 - Infinite resistance
 - device senses force but does not perceptibly move
 - force is mapped to rate control of the cursor (cursor velocity) or to absolute position (force applied)
- Isotonic
 - Device/handle is moved
 - displacement devices, free moving devices or unloaded devices
 - constant resistance
 - displacement of device is mapped to displacement of the cursor
- Elastic
 - Device/handle is moved
 - Device's resistive force increases with displacement
 - Device can sense displacement or force
 - Force/displacement is mapped to rate control of the cursor (typical) or to absolute position

Transfer function (2DOF, 6DOF)

- Position control

- device displacement is mapped/scaled to position (typically for free moving/isotonic devices, also for elastic devices)
- absolute force is mapped/scaled to position (for isometric or elastic devices)
- Rate control
 - force or displacement is mapped onto cursor velocity
 - Integration of input over time → first order control

Drawback of DOF interaction using a single Pointing device:

- With 2DOF most often time multiplexing is implied!
- One operation at the time (e.g. slider can be only be moved sequentially with the mouse)

Chapter 4 - Scientific Foundations

4.1 What is research?

Research:

- Often, the word is used simply to add weight to an assertion ("Our research shows that ...").
- We demand evidence, evidence meeting a standard of credibility such that the statement is beyond dispute. Providing such credibility is the goal of research.
- Three definitions:
 1. Careful or diligent search.
 2. Collecting information about a particular subject.
 3. Investigation or experimentation aimed at the discovery and interpretation of facts and revision of accepted theories or laws in light of new facts.
 - In HCI, an experiment is sometimes called a user study.
 - A formal and standardized methodology is generally preferred because it brings consistency to a body of work and facilitates the

review and comparison of research.

- Facts: evidence we seek in experimental research. Context is also important. Collectively, the facts form an outward sign leading to evidence.
- In HCI research we don't prove things; we gather facts and formulate and test evidence.
- Theory: synonymous with hypothesis, when confirmed through research, a theory becomes a scientifically accepted body of principles that explain phenomena.
- Law (more specific and binding than theory): a relationship or phenomenon that is "invariable under given conditions." Variability is germane to human behavior → laws are of questionable relevance to HCI.
 - Fitts' law: a body of work, originally in human motor behavior, but now widely used in HCI. He proposed a model of human motor behavior. It is a law only in that other researchers took up the label as a celebration of the generality and importance of Fitts' seminal work.
- Discovery: that's what we do, look for, or discover, things that are new and useful.
- Interpretation and revision are central to research. Today's research builds on what is already known or assumed. We interpret what is known; we revise and extend through discovery.

4.1.1 Research must be published

Until it is published, knowledge gained through research cannot achieve its critical purpose, to extend, refine, or revise the existing body of knowledge in the field.

- Research results are "written up," submitted, and reviewed for their integrity, relevance, and contribution. Peer review can either accept or deny to publish work.

Bona fide inventions: real and original inventions made with honest intent.

Patenting is an option to profit from an invention. If the application is successful, the patent is granted and the inventor or company thereafter owns the rights to the invention. A patent is a publication.

4.1.2 Citations, references, impact

Citations connect research papers to other research papers.

Purposes of citations:

- Supporting intellectual honesty.
- Important to back up assertions that are otherwise questionable.
- In the Results section of a research paper, citations are used to compare the current results with those from earlier research.

If many researchers cite a single paper, there is a good chance the work described in the cited paper is both of high quality and significant to the field.

Impact of a researcher's publication record measured by the **H-index**: quantifies both research productivity (number of publications) and overall impact of a body of work (number of citations).

4.1.3 Research must be reproducible

Research that cannot be replicated is useless. Achieving an expected standard of reproducibility, or repeatability, is therefore crucial.

- Standardized methodology enforces a process for conducting and writing about the research that ensures sufficient detail is included to allow the results to be replicated. A guidance can be found in the methodology.
- A researcher questioning a result can redo the experiment and therefore verify or refute the result.

4.1.4 Research versus engineering versus design

Engineers and designers: build things. They create products that strive to bring together the best in form (design emphasis) and function (engineering emphasis).

- Trade-off between form and function.

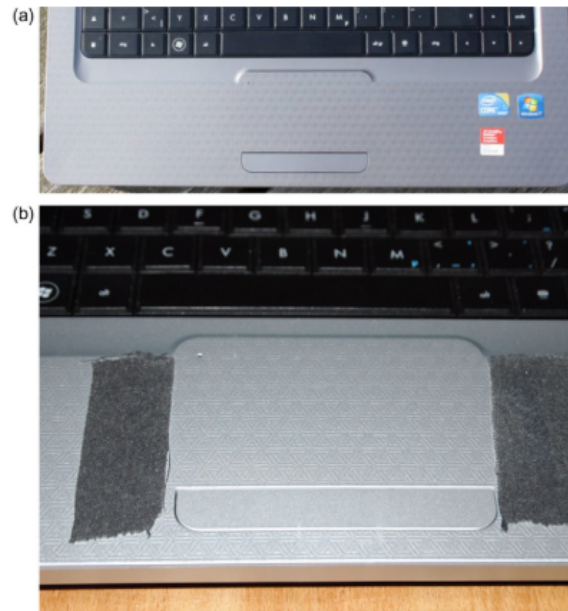


FIGURE 4.2

Form trumping function: (a) Notebook computer. (b) Duct tape provides tactile feedback indicating the edge of the touchpad.

Trumping function: a feature that takes priority over others, deciding the final outcome.

Research: narrowly focused. Small ideas are conceived of, prototyped, tested, then advanced or discarded. New ideas build on previous ideas and good ideas are refined into the building blocks that find their way into products. But research questions are generally small in scope.

Engineers and designers also work with prototypes, but the prototype is used as part of product development. A researcher's prototype is an early mock-up of an idea, and is unlikely to directly appear in a product.

- The more "finished" a prototype seems, the less likely its creators will be to pay attention to and profit from feedback.

Timeline of research is different than that of design and engineering. Research precedes engineering and design and it is slower.

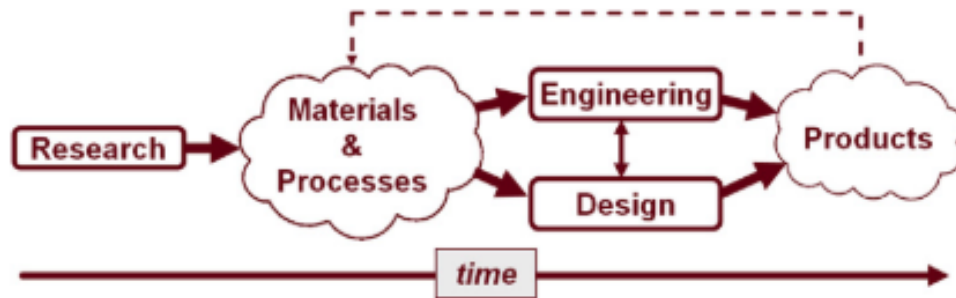


FIGURE 4.3

Timeline for research, engineering, and design.

Researchers provide the raw materials and processes engineers and designers work with.

4.2 What is empirical research?

Empirical:

- Originating in or based on observation or experience.
- Relying on experience or observation alone, often without regard for system and theory.
- Capable of being verified or disproved by observation or experiment.

4.3 Research Methods

Three common approaches or methods:

1. Observational method
2. Experimental method
3. Correlational method

4.3.1 observational method

Observation through interviews, field investigations, contextual inquiries, case studies, field studies, focus groups, think aloud protocols, storytelling, walkthroughs, cultural probes, etc.

- Tends to be qualitative rather than quantitative.

- Achieve relevance while sacrificing precision.
 - Real world phenomena are high in relevance, but lack the precision available in controlled laboratory experiments.
- Concerned with discovering and explaining the reasons underlying human behavior.
- Observations are more likely to involve note-taking, photographs, videos, or audio recordings rather than measurement.
 - Measurements, if gathered, use categorical data or simple counts of phenomena.
- Tend to examine and record the quality of interaction rather than quantifiable human performance.

4.3.2 Experimental method

Controlled experiments conducted in laboratory settings. Can both gather new knowledge and study existing knowledge.

- Less relevance, more precision due to control since over setting (extraneous factors are reduced).
- Requires at least two variables:
 - **Manipulated variable (or independent variable or factor)**
 - **Response variable (or dependent variable)**, a property of human behavior that is observable, quantifiable, and measurable.
- Controlled experiment → at least two configurations required. Comparison is germane to the experimental method.
 - A "**usability evaluation**," on the other hand, usually involves assessing a single user interface for strengths and weaknesses (not experimental).
- We engage observational methods by soliciting comments, thoughts, and opinions from participants.
- Often allows a powerful form of conclusion to be drawn from the data and analyses. The relationship between the independent variable and the

dependent variable is one of cause and effect. This is not possible with the observational or the correlational method.

4.3.3 Correlational method

Involves looking for relationships between variables.

- Characterized by quantification since the magnitude of variables must be ascertained (e.g., age, income, number of privacy settings).
 - For nominal-scale variables, categories are established (e.g., personality type, gender).
- The data may be collected through a variety of methods.
- Often accompany experimental methods if questionnaires are included in the experimental procedure.
- Balance between relevance and precision.
- Data obtained using correlational methods are circumstantial, not causal.

4.4 observe and measure

4.4.1 Observation

The starting point for empirical research in HCI is to observe humans interacting with computers. Two possibilities to make observations:

- Another human (experimenter or investigator) is the observer, measurements are collected manually.
 - Limitation: accuracy.
 - Useful for pilot (or preliminary) testing.
- An apparatus (e.g. a computer) is the observer.
 - Sometimes it is not possible to embed data collection software in the apparatus and some behaviors of interest are difficult to detect.

4.4.2 Measurement scales

With measurement, anecdotes turn to empirical evidence.

Four scales of measurement:

1. Nominal
2. Ordinal
3. Interval
4. Ratio

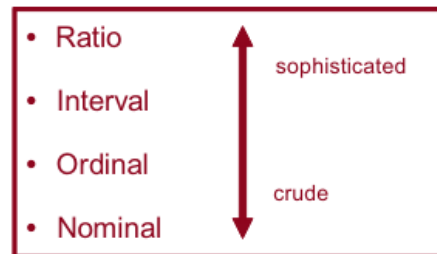


FIGURE 4.4

Scales of measurement: nominal, ordinal, interval, and ratio. Nominal measurements are considered simple, while ratio measurements are sophisticated.

Which one is most and least sophisticated follows from the types of computations possible with each measurement.

4.4.3 Nominal

A measurement on the nominal scale (or categorical data):

- Arbitrarily assigning a code to an attribute or a category.
- The code does not need to be a number (it can be).
- Mathematical manipulations on nominal data are meaningless.
- Nominal data identify mutually exclusive categories.
 - The only relationship that holds is equivalence, which exists between entities in the same class.
- Often used with frequencies or counts (the number of occurrences of each attribute).
 - The attribute is categorical, the count is a ratio-scale measurement.

4.4.4 Ordinal data

Ordinal scale measurements:

- Provide an order or ranking to an attribute. The attribute can be any characteristic or circumstance of interest.
- Limitation: the interval is not intrinsically equal between successive points on the scale.
- Slightly more sophisticated than nominal data since comparisons of greater than or less than are possible. It is not valid to compute the mean of ordinal data.
- Likert scale is ordinal data, but is treated as interval data in statistical analysis (although there's some debate about this).

4.4.5 Interval data

Interval data:

- Equal distances between adjacent values.
- No absolute zero.
- Meaningful to compute the mean. Ratios of interval data are not meaningful (e.g. one cannot say that 20°C is twice as warm as 10°C).
- Questionnaires where a response on a linear scale is solicited (e.g. a Likert Scale, where verbal responses are given a numeric code).
 - There is some disagreement among researchers on the assumption of equal gradations between these items. Do respondents perceive the difference between strongly disagree and mildly disagree the same as the difference between mildly disagree and neutral?
 - Attaching verbal tags to numbers is likely to bring personal interpretations. There is evidence that respondents perceive items at the extremes of the scale as farther apart than items in the center.
 - One solution: instruct respondents to interpret the items as equally spaced.

4.4.6 Ratio data

Ratio-scale measurements:

- Most sophisticated of the four scales of measurement.
- Have an absolute zero.
- Lots of calculations possible.
- In HCI, the most common ratio-scale measurement is time.
- Count is a ratio
 - The expressive nature of a count is improved through normalization (expressing the value as a count per something).
 - Improves comparisons and adds meaning.

4.5 Research questions

Often the questions pertain to the relationship between two variables, where one variable is a circumstance or condition that is manipulated (an interface property) and the other is an observed and measured behavioral response (task performance).

Difficulty: people exhibit variability in their actions.

Research questions need to be testable.

- Avoid "any good" or "better than". How does one measure "better than"?
- Preferable to address qualities that are more easily observed and measured.
- Using alternative designs so comparisons are possible.
- A testable and good research question:
 - Is the measured entry speed (in words per minute) higher for the new technique than for a QSK after one hour of use?
 - Speaks very specifically to entry speed measured in words per minute, to a comparison between two methods, and to a criterion for practice.

4.6 Internal validity and external validity

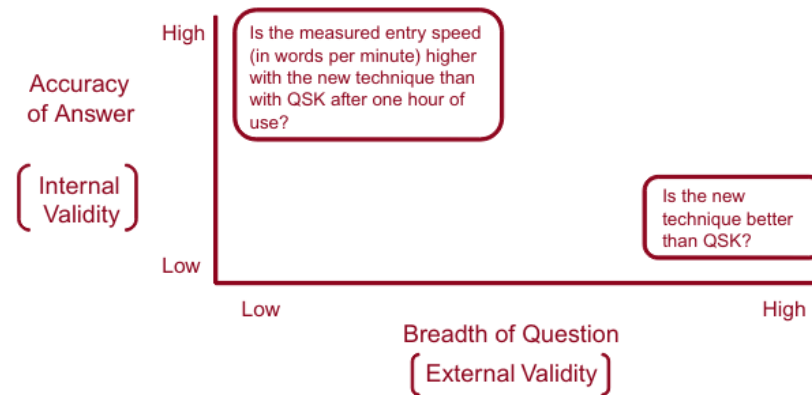


FIGURE 4.8

Graphical comparison of Internal Validity and External Validity.

Internal validity: the extent to which an effect observed is due to the test conditions.

- High internal validity means the effect observed really exists.

External validity: the extent to which experimental results are generalizable to other people and other situations.

- Participants involved were representative of a larger intended population. In experimental research, random sampling is important for generalizability.
 - **Generalizable** to “other situations”: the experimental environment and procedures were representative of real world situations where the interface or technique will be used.

Trade-off between internal and external validity.

- No universal remedy for the tension between internal and external validity. One must acknowledge the limitations.
 - We can pose multiple narrow (testable) questions that cover the range of outcomes influencing the broader (untestable) questions, will increase both internal and external validity.

Ecological validity: refers to the methodology (using materials, tasks, and situations typical of the real world).

4.7 Comparative evaluations

A comparative study leads to more meaningful and insightful results. Variations in the new design, an established design (a baseline condition), or some combination of the two can be compared.

A controlled experiment must include at least one independent variable and the independent variable must have at least two levels or test conditions.

Benefits of including an established design as a baseline condition:

1. The baseline condition serves as a check on the methodology. Baseline conditions are well traveled in the research literature, so results in a new experiment are expected to align with previous results.
2. The baseline condition allows results to be compared with other studies. This facilitates comparisons between the two user studies. Furthermore, conclusions can often be expressed in terms of the known baseline condition.

Study: participants of a comparative study were more critical towards interfaces, and were less prone to inflate their subjective ratings.

4.8 Relationships: circumstantial and causal

If a controlled experiment is done properly, we can say that the condition manipulated in the experiment caused the changes in the human responses that were observed and measured. This is a **causal relationship**.

Finding a relationship does not necessarily mean a causal relationship exists. Many relationships are circumstantial. They exist, and they can be observed, measured, and quantified. Circumstantial relationships can be useful, they are often the first step in further research.

Causal relationships emerge from controlled experiments. Requirement:

- Participants are selected randomly from a population and are randomly assigned to test conditions. Sometimes participants are balanced into groups where the participants in each group are screened so that the groups are equal in terms of other relevant attributes.
 - So that differences between groups are really because of the test conditions and not individual differences.

Cause and effect conclusions are not possible:

- If the variable manipulated is a naturally occurring attribute of participants.
 - It is not possible to avoid confounding variables. It is not possible to know whether the experimental effect was due to the independent variable or due to the confounding variable.

Chapter 5 - Designing HCI Experiments

One way to think about experiment design is through a signal and noise metaphor:

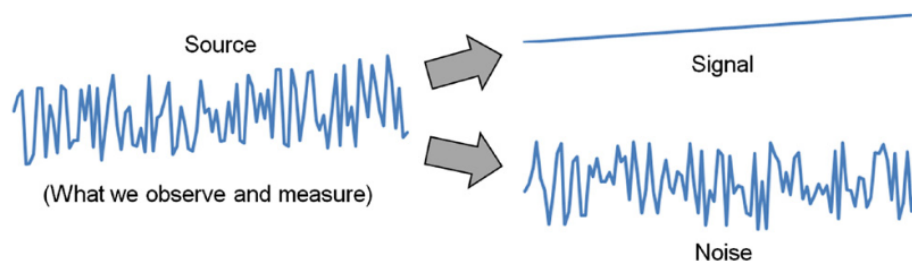


FIGURE 5.1

Signal-to-noise conceptualization of experiment design.

Signal: a variable of interest.

Noise: everything else, the random influences.

5.1 What methodology?

Method or methodology: the way an experiment is designed and carried out.

Involves:

- Participants
- Materials or apparatus
- (order of) tasks
- procedure for preparing the participants
- Variables
- Data collected and analyzed, etc.

Methodology is the bedrock (:) of science. If the methodology is weak or flawed, there is no science forthcoming. What remains is little else than commentary.

Use of a standardized methodology strengthens experimental research. The flip side is that an ad hoc, or made-up, methodology weakens research.

5.2 Ethics approval

Since HCI research involves humans, “researchers must respect the safety, welfare, and dignity of human participants in their research and treat them equally and fairly.”

The approval process is governed by the institution or funding agency overseeing the research.

Typically, the review committee serves to ensure a number of ethical guidelines are acknowledged and adhered to. These include the right of the participant to be informed of:

- The nature of the research (hypotheses, goals and objectives, etc.)
- The research methodology (e.g., medical procedures, questionnaires, participant observation, etc.)
- Any risks or benefits
- The right not to participate, not to answer any questions, and/or to terminate participation at any time without prejudice
- The right to anonymity and confidentiality

5.3 Experiment design

Experiment design: the process of bringing together all the pieces necessary to test hypotheses on a user interface or interaction technique. Involves deciding on and defining which variables to use, what tasks and procedure to use, how many participants to use and how to solicit them, etc.

Thinking about experimental variables forces us to craft narrow and testable questions. The two most important experimental variables are independent variables and dependent variables.

A testable research question inherently expresses the relationship between an independent variable and a dependent variable.

5.4 Independent variables

Independent variable (or factor): a circumstance or characteristic that is manipulated or systematically controlled to a change in a human response while the user is interacting with a computer.

- Experiments designed with independent variables are often called **factorial experiments**. The variable is manipulated across multiple (at least two) levels of the circumstance or characteristic. The variable is “independent” because it is independent of participant behavior.
- The variable manipulated is typically a nominal-scale attribute, often related to a property of an interface.
- Independent variables are anything that might affect users’ proficiency in using a computer system.
- An independent variable can be many things besides an attribute of a computer system. It can be characteristics of humans or an environmental circumstance. Tips:
 1. When formulating an independent variable, express it both in terms of the circumstance or characteristic itself as well as the levels of the circumstance or characteristic chosen for testing. (The levels of an independent variable are often called test conditions.)
 2. Once the name of the independent variable and the names of the levels are decided, stick with these terms consistently throughout a paper.

With every additional independent variable, more effects exist between the variables. A design with two independent variables includes two main effects and one interaction effect, for a total of three effects. With too many variables and thus a lot of effects, it is difficult to find meaningful interpretations for all the effects. Furthermore, variability in the human responses is added with each independent variable. Interaction effects that are three-way or higher are extremely difficult to interpret and are best avoided. Limit it to one or two, three at most.

5.5 Dependent variables

Dependent variable: measured human behavior. In HCI the most common dependent variables relate to speed (e.g. task completion time) and accuracy.

- Accuracy is often reported as the percentage of trials or other actions performed correctly or incorrectly. In the latter case, accuracy is called errors or **error rate**.
- The measurements depend on what the participant does.
- Any observable, measurable aspect of human behavior is a potential dependent variable. Provided the behavior has the ability to differentiate performance between two test conditions in a way that might shed light on the strengths or weaknesses of one condition over another, then it is a legitimate dependent variable.
- It is essential to clearly define all dependent variables to ensure the research can be replicated.
- It is often helpful to name the variable separately from its units.
- It is important to consider how the measurements are gathered and the data collected, organized, and stored. The most efficient method is to design the experimental software to gather the measurements. The data should be organized and stored in a manner that facilitates followup analyses. To facilitate follow-up analyses, the data should also include codes to identify the participants and test conditions.

5.6 Other variables

There are three other variables:

1. Control
2. Random
3. Confounding

5.6.1 Control variables

There are many circumstances or factors that (a) might influence a dependent variable but (b) are not under investigation. These need to be accommodated in

some manner. One way is to control them: to treat them as control variables. Controlling them means that they are fixed at a nominal setting during the experiment so they don't interfere. But they might interfere if set at an extreme value. Having lots of control variables reduces the variability in the measured behaviors but yields results that are less generalizable.

5.6.2 Random variables

Instead of controlling all circumstances or factors, some might be allowed to vary randomly. Such circumstances are random variables. More variability is introduced in the measures, but there is a benefit since results are more generalizable.

Variable	Advantage	Disadvantage
Random	Improves external validity by using a variety of situations and people.	Compromises internal validity by introducing additional variability in the measured behaviours.
Control	Improves internal validity since variability due to a controlled circumstance is eliminated	Compromises external validity by limiting responses to specific situations and people.

FIGURE 5.4

Relationship between random and control variables and internal and external validity.

5.6.3 Confounding variables

Confounding variable: any circumstance or condition that changes systematically with an independent variable. Is the effect observed due to the independent variable or to the confounding variable? Researchers must attune to the possible presence of a confounding variable and eliminate it, adjust for it, or consider it in some way. Otherwise, the effects observed may be incorrectly interpreted.

5.7 Task and procedure

When participants are given a test condition, they are asked to do a task while their performance is measured. Later, they are given a different test condition and asked to do the task again.

Two objectives in designing a good task:

1. A task that is similar to actual or expected usage will improve the external validity of the research.

- a. Downside: the more representative the task, the more likely the task is to include behaviors not directly related to the interface or interaction method under test. Such behaviors are likely to compromise the ability of the task to discriminate among the test conditions.
 - b. Downside: decrease in internal validity. Furthermore, the additional variation may bring forth a non-significant statistical result.
2. A good task is one that can discriminate the test conditions. A good task must attune to the points of differentiation in order to elicit behavioral responses that expose benefits or problems among the test conditions. This should surface as a difference in the measured responses across the test conditions. A difference might occur if the interfaces or interaction techniques are sufficiently distinct in the way the task is performed.

The best task is one that is natural yet focuses on the core aspects of the interaction: the points of differentiation between the test conditions. Points of similarity, while true to actual usage, introduce variability.

The experimental procedure includes the task but also the instructions, demonstration, practice or questionnaires. The procedure encompasses everything that the participant did or was exposed to.

5.8 Participants

Applying results to people other than those who were tested is possible; however, two conditions are required:

1. The people tested must be members of the same population of people to whom results are assumed to hold.
2. A sufficient number of participants must be tested. This requirement has more to do with statistical testing than with the similarity of participants to the population.

The variability in the measurements affects the likelihood of obtaining statistically significant results. Increasing the number of participants (large n) increases the likelihood of achieving statistically significant results.

Which n ? → use the same number of participants as in similar research.

- A large n can cause statistically significant results for a difference of no practical significance.
- If not enough participants are used, statistical significance may fail to appear.
- In usability evaluations, there is evidence that about five participants (often usability experts) are sufficient to expose about 80 percent of the usability problems.

When referring specifically to the experiment, use the term participants. General comments on the topic or conclusions drawn may use other terms (e.g. users).

Ideally, participants are drawn at random from a population.

- Convenience sampling compromises the external validity of the research, since the true population is somewhat narrower than the desired population.

Questionnaires are used to collect demographic data and other information relevant to the research.

Depending on the agency or institution overseeing the research, participants are usually required to sign a consent form prior to testing. The goal is to ensure participants know that their participation is voluntary, that they will incur no physical or psychological harm, that they can withdraw at any time, and that their privacy, anonymity, and confidentiality will be protected.

5.9 Questionnaire design

Two purposes of questionnaires:

1. Gather information on demographics and experience with related technology.
2. To solicit participants' opinions on the devices or interaction tasks with which they are tested.

Close-ended questions:

Do you use a GPS device while driving? ☐ yes ☐ no

Which browser do you use?

☐ Mozilla *Firefox* ☐ Google *Chrome*
☐ Microsoft *IE* ☐ Other (_____)

Open-ended question:

Which browser do you use? _____

Closed-ended questions simplify follow-on analyses, since it is straightforward to tally counts of responses.

Age can be solicited as an open-ended ratio-scale response, as seen here:

Please indicate your age: _____

Collected in this manner, the mean and standard deviation are easily calculated. Ratio-scale responses are also useful in looking for relationships in data.

Age can also be solicited as an ordinal response, as in this example:

Please indicate your age.

☐ < 20 ☐ 20-29 ☐ 30-39
☐ 40-49 ☐ 50-59 ☐ 60+

In this case, the counts in each category are tabulated. Such data are particularly useful if there is a large number of respondents. However, ordinal data are inherently lower quality than ratio-scale data, since it is not possible to compute the mean or standard deviation.

Questionnaires are also used at the end of an experiment to obtain participants' opinions and feelings about the interfaces or interaction techniques. Items are often formatted using a Likert scale to facilitate summarizing and analyzing the

responses.

Example:

Eye fatigue:						
1	2	3	4	5	6	7
Very high						Very low

Note that the preferred response is 7, whereas the preferred response in the NASA-TLX example is 1. In the event the mean is computed over several response items, it is important that the items are consistently constructed.

5.10 Within-subjects and between-subjects

If each participant is tested on each level, the assignment is **within-subjects**.

Within-subjects is also called repeated measures.

- More testing needed for each participant.
- The variance due to participants' predispositions will be approximately the same across test conditions.
- It is not necessary to balance groups of participants, because there is only one group.

If each participant is tested on only one level, the assignment is **between-subjects**. For a between-subjects design, a separate group of participants is used for each test condition.

- More participants are needed.
- More variability due to inherent differences between participants.
- Balancing is needed to ensure the groups are more or less equal in terms of characteristics that might introduce bias in the measurements.

Sometimes a factor must be between-subjects (e.g. one group male and one group female). Sometimes a factor must be within-subjects (e.g. measuring results before and after practice).

HCI tends to favor within-subjects designs over between-subjects designs. However, there is an advantage to a between-subjects design, it avoids **interference** (conflict that arises when a participant is exposed to one test condition and then switches to another test condition) between test conditions.

In many areas of research, within-subjects designs are rarely used.

For an experiment with two factors it is possible to assign the levels of one factor within-subjects and the levels of the other factor between-subjects. This is a **mixed design**.

5.11 Order effects, counterbalancing, and latin squares

When the levels of test conditions are assigned within-subjects, interference between the test conditions may result due to the order of testing.

- It is possible that participants' performance will improve as they progress from one test condition to the next. They become familiar with the apparatus and procedure, and they are learning to do the task more effectively. Practice, then, is a confounding variable. This is referred to as a **practice effect** or a **learning effect**.
- It is also possible that performance will worsen on conditions that follow other conditions. This may follow from mental or physical fatigue, a **fatigue effect**.
- In a general sense, the phenomenon is an **order effect** or **sequence effect** (can either improve or degrade performance).

Counterbalancing is the most common method of compensating for an order effect.

- **Counterbalancing:** the compensatory ordering of test conditions to offset practice effects.

A **Latin square** is an $n \times n$ table filled with n different symbols (e.g., A, B, C, and so on) positioned such that each symbol occurs exactly once in each row and each column.

(a)

A	B
B	A

(b)

A	B	C
B	C	A
C	A	B

(c)

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

(d)

A	B	C	D	E
B	C	D	E	A
C	D	E	A	B
D	E	A	B	C
E	A	B	C	D

FIGURE 5.7

Latin squares: (a) 2×2 . (b) 3×3 . (c) 4×4 . (d) 5×5 .

A deficiency in Latin squares of order 3 and higher is that conditions precede and follow other conditions an unequal number of times. A solution to this is a **balanced Latin square**, which can be constructed for even-order tables.

(a)

A	B	D	C
B	C	A	D
C	D	B	A
D	A	C	B

(b)

A	B	F	C	E	D
B	C	A	D	F	E
C	D	B	E	A	F
D	E	C	F	B	A
E	F	D	A	C	B
F	A	E	B	D	C

FIGURE 5.8

Balanced Latin squares where each condition precedes and follows other conditions an equal number of times: (a) 4×4 . (b) 6×6 .

When designing a within-subjects counterbalanced experiment, the number of levels of the factor must divide equally into the number of participants.

There is a potential problem for the 3×3 Latin square. B follows A twice, but A follows B only once. So there is an imbalance. One solution in this case is to counterbalance by using all sequences.

A	B	C
A	C	B
B	C	A
B	A	C
C	A	B
C	B	A

FIGURE 5.11

Counterbalancing an odd number of conditions using all ($n!$) combinations.

Another way to offset learning effects is to randomize the order of conditions. This is most appropriate where (a) the task is very brief, (b) there are many repetitions

of the task, and (c) there are many test conditions.

5.12 Group effects and asymmetric skill transfer

If the learning effect is the same from condition to condition in a within-subjects design, then the group means on a dependent variable should be approximately equal. However, there are occasions where different effects appear for one order (e.g., $A \rightarrow B$) compared to another (e.g., $B \rightarrow A$). In such cases there may be a **group effect** (differences across groups in the mean scores on a dependent variable).

- Counterbalancing did not work.
- A group effect is typically due to **asymmetric skill transfer**. Solutions:
 - Use a between-subjects design.
 - Have participants practice on a condition prior to data collection. The practice trials seek to overcome the benefit of practice in the earlier condition, so that the measured performance accurately reflects the inherent properties of the test condition.

5.13 Longitudinal studies

When wanting to measure learning or acquisition of skill, the experimental procedure involves testing users over a prolonged period while their improvement is measured. An experimental evaluation where participants practice over a prolonged period is called a **longitudinal study**.

- "Amount of practice" is an independent variable. Each unit of testing is a level of the independent variable.

Performance trends in longitudinal studies, are often accompanied by an equation and best-fitting curve demonstrating the power law of learning.

In many situations, the goal of a longitudinal study is to compare the viability of a new technique against current practice. If users are experienced with a current interaction technique, then relatively poorer initial performance is expected with the new technique. Learning a new technique bears a cost, since performance is initially superior with the current technique. However, after the crossover point is reached, the new technique provides a benefit, since performance is superior compared to current practice.

It is often expensive to change the current design and users are change-adverse.

5.14 Running the experiment

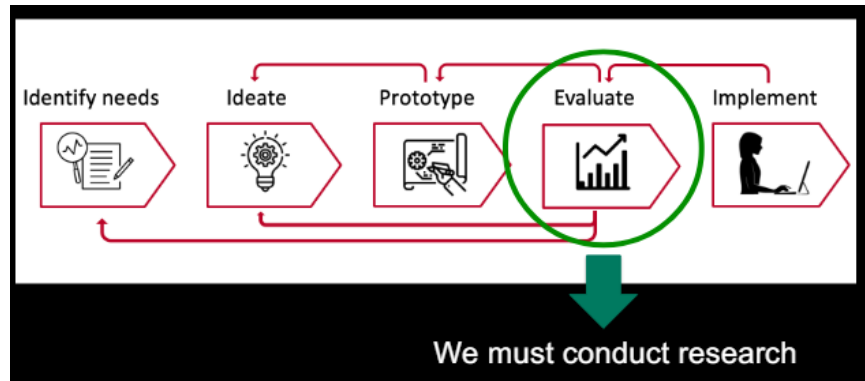
It is always useful to have a pilot test with one or two participants. This will help smooth out the protocol for briefing and preparing the participants and checks the time it takes.

Conducting an experiment:

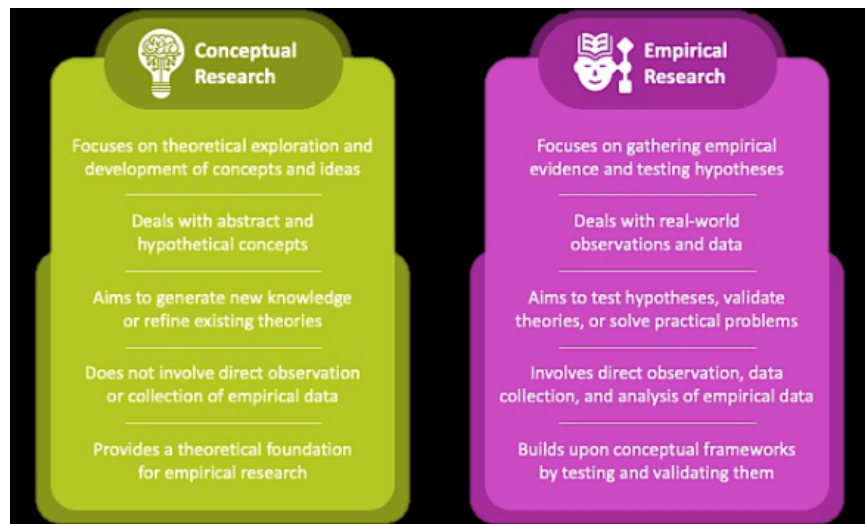
- Greeting, introducing the experiment, lets participants sign a consent form.
 - For most interaction tasks, the participant is expected to proceed quickly and accurately. Performing tasks reasonably quick and with high accuracy, but at a rate comfortable to the individual, is usually the goal.
 - Instructions must be carefully considered and must be given to all participants in the same manner. If a participant asks for clarification, caution must be exercised in elaborating on the instructions. Any additional explanation that might motivate a participant to act differently from other participants is to be avoided.
 - It is important that the experimenter portrays himself or herself as neutral.
- Questionnaire to gather demographic data and information on the participants' related experience.
- Apparatus is revealed, the task explained and demonstrated. Practice trials are allowed, as appropriate.

Additional notes lecture week 3

Research



Research: investigation or experimentation aimed at the discovery and interpretation of facts and revision of accepted theories or laws in light of new facts.

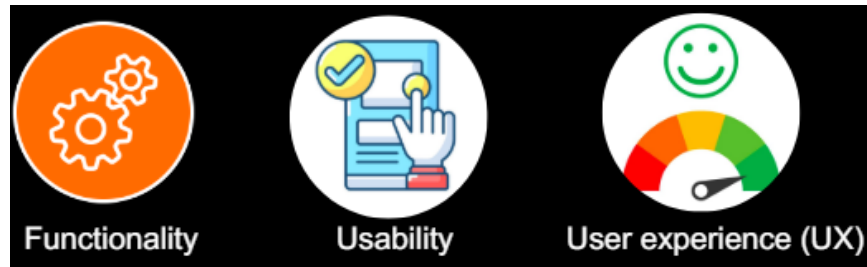


We conduct research:

- Because scientists want to understand cause and effect.
- To make predictions.
- To test usability and user experience.

Evaluation in HCI has three goals:

1. To identify any specific problems or errors in the design.
2. To assess the usability and accessibility of the design.
3. To assess users' experience with the design.



Research must be:

- Reproducible (rigorous method)
- Peer reviewed and published (detailed report)
- Builds on previous scientific evidence (references & citations)



Isolating the cause and effect

- Participants rated the system easy to use, because
 - they actually found the system easy to use

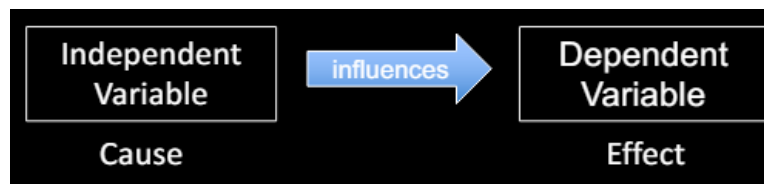
- they want to support you in your research
- they were overwhelmed by the system's novelty
- the Dutch football team won the world cup yesterday
- ...
- Mere observation will NOT help to find the answer!
- Controlled experiments are more reliable because
 - They can isolate cause and effect!
 - Knowing cause and effect allows informed design

Correlation $\neq$ causation

RQ and Hypothesis

Hypothesis: a prediction about the relationship between two or more variables.

RQ (research question): a specific question regarding the assumed relationship that the study tries to answer.



- In HCI research, RQs are often about new or existing user interface or interaction technique.
- Variables can be any attribute or property of humans, systems or environments.

Hypotheses must be testable and measurable

- Keyboard X is **better than** keyboard Y.
Increases typing speed
- Adding element XX **improves usability** of the interface.
Reduces errors / increases user performance
- Adding auditory feedback in cars **improves situational awareness** in drivers.
Faster response to events in the driving simulator (e.g., braking car / pedestrian crosses road)

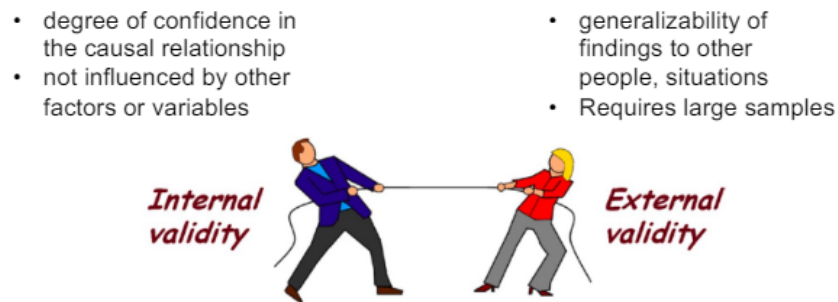


FIGURE 4.9

There is tension between internal validity and external validity. Improving one comes at the expense of the other.

Measurements

- Literature review
 - Consensus in a scientific community
 - Validated scales and questionnaires
 - Results are comparable
- If does not exist, define it for your experiment
 - "For this experiment, we define driving experience as years of having a driver's license."
 - Less valid but quantified
 - Harder to compare to other results

Evaluation

Consensus in HCI community

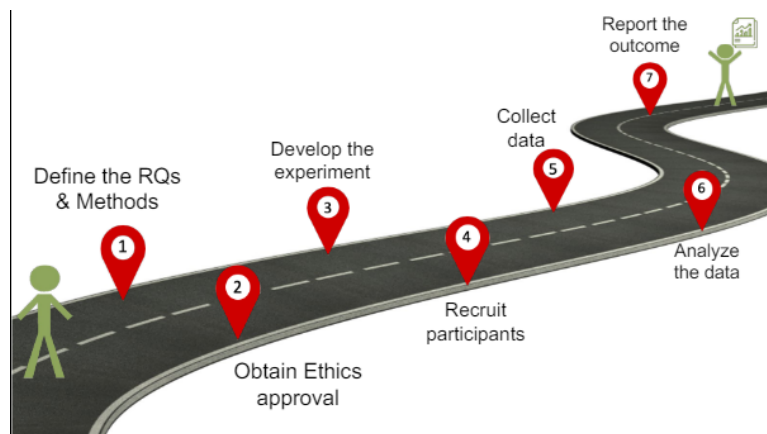
- Efficiency – Task completion time
- Accuracy – error rate
- Learnability – repeated measure of reaction time or performance
- ...

Standardized questionnaires

- Nasa-TLX (task load)
- AttrakDiff (attractiveness of UI)
- SUS (Usability)
- UTAUT (acceptance and use of UI)

Designing HCI Experiments

Define your methodology



Ethical approval

- The approval process is normally governed by the institution where the research is conducted (Institutional Review Board (IRB), Ethics Review Committee (ERC), etc.)

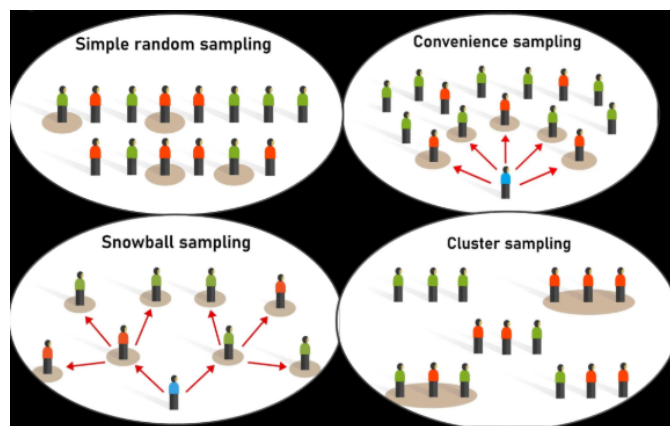
Ethics guidelines

- Participants have the right to know:
 - The nature of the research (hypotheses, goals and objectives, etc.)

- The research methodology (e.g., medical procedures, questionnaires, participant observation, etc.)
- Any risks or benefits
- The right not to participate, not to answer any questions, and/or to terminate participation at any time without penalty
- The right to anonymity and confidentiality

Define methodology

- Method or methodology: the way an experiment is designed and carried out.
 - Participants
 - Take a sample that reflects the population.



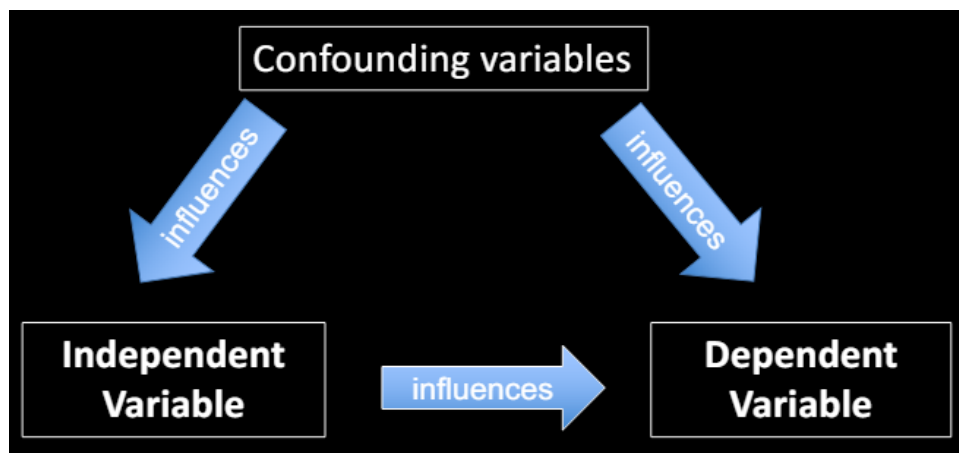
- More participants → higher statistical validity, higher costs
- For your research project recruit 10-12 participants per condition (our TA said at least 5 though)
- High variability in means requires more participants
- Reporting participants:
 - Sampling method
 - Sample size
 - Average age and gender split
 - Recruitment process

- Participants' background
 - Other things if needed for specific study
- Materials or apparatus (hardware and software)
- User tasks
- Experimental design and conditions (IV)
 - Between-Group Design
 - Advantages
 - Simplicity
 - Less chance of practice or fatigue effects
 - Useful when it is impossible for an individual to participate in all conditions
 - Disadvantages
 - Expense (time, effort, and number of participants)
 - Individual differences (different responses to experimental manipulations)
 - Within-Group Design
 - Advantages
 - Economy
 - Sensitiveness
 - Cancelling out individual differences
 - Disadvantages
 - Carry-over effects from previous conditions
 - Conditions need to be reversible
 - Consider the novelty effect
- Procedure (the order of tasks/conditions)

- Evaluation (DV)
- Data analysis

Variables

- Independent variables
- Dependent Variables
- Uncontrolled variables (confounding variables)



How to identify confound variables?

- Think about possibilities
 - What else could influence the results?
- Review the literature
 - What is already known?
- Experience
 - Talk to colleagues and supervisors

How to control confound variables?

- Randomize
- Record
 - Analyze it later as co-variable

If confound variables are not controlled, the recorded data can be unusable.

In within-subjects studies, you must randomize the order of conditions to counteract repeated exposure to the same interface and minimize the learning effect.

Wizard of Oz thing

"Wizard of Oz is a rapid prototyping method for systems costly to build or requiring new technology. A human "Wizard" simulates the system's intelligence and interacts with the user through a real or mock computer interface."

- An invisible 'wizard' controlling parts of the functionality
- We only implement the easy parts but leave the hard part to the human operator
- Provides the user with the experience without extensive implementation effort for the prototype
- Typical areas
 - Speech recognition
 - Speech synthesis
 - Reasoning
 - Computer vision
- Wizard of Oz
 - Implemented only the "soft" part
 - Using human operators for the hard parts
- Advantages
 - Receive early feedback
 - Compare with alternatives
- Disadvantages
 - Even implementing the soft parts requires effort

- Unclear if operator's performance can ever be reached
- Performance depends on the operator

Chapter 6 - Hypothesis Testing

Parametric tests: operate on data from a probability distribution, such as the normal distribution or the t-distribution.

- Have a greater ability (power) to draw conclusions in the statistical tests undertaken.

Non-parametric tests: are "distribution free," they make no assumption about the distribution of the underlying data.

- Applicable to any scale of data (though most commonly used for nominal or ordinal data).

Measurement Scale	Defining Relations	Examples of Appropriate Statistics	Appropriate Statistical Tests
Nominal	• Equivalence	• Mode • Frequency	• Non-parametric tests
Ordinal	• Equivalence • Order	• Median • Percentile	
Interval	• Equivalence • Order • Ratio of intervals	• Mean • Standard deviation	• Parametric tests • Non-parametric tests
Ratio	• Equivalence • Order • Ratio of intervals • Ratio of values	• Geometric mean • Coefficient of variation	

FIGURE 6.1

Measurement scales of data, properties of data, and appropriate statistical tests.

6.1 Analysis of variance

An **ANOVA (or F-test)** is the main statistical procedure for hypothesis testing in factorial experiments.

- The majority of HCI research papers that describe experiments include the results of an ANOVA giving:
 - F-statistic
 - Degrees of freedom

- P-value.
- An ANOVA determines if an independent variable had a significant impact on a dependent variable.
 - The ANOVA determines this by partitioning the variance in the observations on a dependent variable into components attributable (the test conditions and the participants) to the variance. If the variance across levels of a factor is large relative to all the remaining variance (other factors and the participants), it is often possible to conclude that the distinguishing properties of the factor (features in the test conditions) caused the observed differences in the measurements on the dependent variable. If so, the factor is said to have had a statistically significant effect on the dependent variable.

6.1.1 Why analyze the variance?

The statistical question pursued is not a question, it is a statement known as the **null hypothesis**, which assumes that there is no difference between two conditions.

In most cases, the researcher seeks to reject the null hypothesis. A statistically significant result means there is little likelihood the null hypothesis is true. The conclusion in such case is that (a) there is a difference in the means, (b) the difference is statistically significant, and (c) the difference was caused by distinguishable properties in the test conditions.

If the difference is not significant, it implies that the difference is through chance.

Variance: quantifies the spread or dispersion of a set of data points around their mean value, indicating how much the values differ from the average. It is calculated as the average of the squared differences between each data point and the mean (square of the standard deviation).

ANOVA Table for Task Completion Time (s)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	9	5.080	.564				
Method	1	4.232	4.232	9.796	.0121	9.796	.804
Method * Subject	9	3.888	.432				

FIGURE 6.4

Analysis of variance table for data in [Figure 6.3a](#).

P-Value: the probability of obtaining the observed data if the null hypothesis is true.

- In experimental research, significance, in a statistical sense, requires p less than .05, or a 1 in 20 probability.

One reason experimental research is so valuable is that conclusions of a cause-and-effect nature are possible.

Other important statistics in the ANOVA are the F statistic (F-Value, the ratio of sample variances) and the degrees of freedom for the main effect (Method) and the residual effect (Method * Subject).

The mean task completion time for Method A was 4.5 s. This was 20.1% less than the mean of 5.5 s observed for Method B. The difference was statistically significant ($F_{1,9} = 9.80, p < .05$).

FIGURE 6.5

Example of how to report the results of an analysis of variance in a research paper.

- P is not precisely cited, but as less than: {.05, .01, .005, .001, .0005, .0001}. So p is cited as $p < .05$ rather than $p = .0121$.
- It includes both the means for the test conditions and an articulation of the difference in the means. The difference in the means is the effect size. The effect size is usually conveyed in HCI research in a statement giving the means and the absolute difference (one second, in this case) or the relative difference.

Formatting the F statistics:

- Placed in parentheses
- Uppercase for F
- Lowercase for p
- Italics for F and p
- Space on both sides of the equal sign
- Space after the comma
- Space on both sides of the less than sign
- Degrees of freedom are subscript, plain, smaller font
- Three or four significant figures for the F statistic
- No zero before the decimal point for the p statistic (because it is constrained between 0 and 1)

Non-significant results are also important to report. It does not imply that the null hypothesis is true. It indicates that the evidence is insufficient to reject the null hypothesis: The null hypothesis remains tenable.

For non-significant results, the ANOVA result is reported in one of two ways.

- If F is less than 1, it is impossible for the effect to be significant, and in this case "ns," for "not significant," is substituted for p.
- If the F statistic is greater than 1 and p is greater than .05, the result does not meet the acceptable threshold for significance and p is reported as $p > .05$.

6.1.2 More than two test conditions

ANOVA Table for Dependent Variable (units)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	15	81.109	5.407				
Test Condition	3	182.172	60.724	4.954	.0047	14.862	.896
Test Condition * Subject	45	551.578	12.257				

FIGURE 6.10

ANOVA for data in [Figure 6.9a](#).

If n is the number of test conditions and m is the number of participants, then the degrees of freedom are $(n - 1)$ for the variance due to Test Condition and $(n - 1)(m - 1)$ for the variance due to Test Condition $\times$ Subject.

6.1.3 Post hoc comparisons

To determine which test conditions differ from which other test conditions, a Scheffé post hoc comparisons test is used.

Scheffe for Dependent Variable (units)
Effect: Test Condition
Significance Level: 5 %

	Mean Diff.	Crit. Diff.	P-Value	
A, B	-.875	3.302	.9003	
A, C	-4.500	3.302	.0032	S
A, D	-1.813	3.302	.4822	
B, C	-3.625	3.302	.0256	S
B, D	-.938	3.302	.8806	
C, D	2.688	3.302	.1520	

FIGURE 6.11

Scheffé post hoc comparisons test for data in Figure 6.9. "S" identifies comparisons that meet the $p < .05$ threshold for significance.

The means are significantly different between test conditions A and C and between test conditions B and C.

6.1.4 Between-subjects designs

The between-subjects factor is identified as an added column of nominal data encoding the levels of the factor. "L" and "R" identify the participants who were left handers versus right handers.

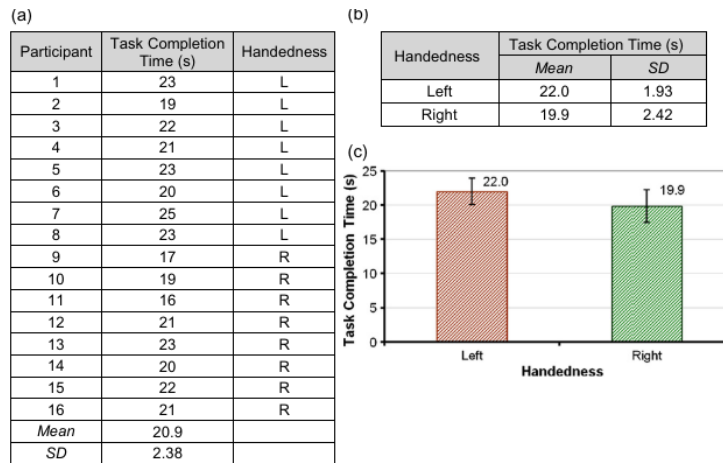


FIGURE 6.12

Experiment with handedness as a between-subjects factor: (a) Data used in the ANOVA. (b) Summary data showing handedness results. (c) Bar chart with error bars showing ± 1 standard deviation about the means.

ANOVA Table for Task Completion Time (s)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Handedness	1	18.063	18.063	3.781	.0722	3.781	.429
Residual	14	66.875	4.777				

FIGURE 6.13

Between-subjects ANOVA for data in Figure 6.12a.

The ANOVA is performed on the 16×2 matrix in the right two columns in Figure 6.12a. The result is shown in Figure 6.13. The degrees of freedom are $(n - 1)$ for the variance due to Handedness and $(m - n)$ for the variance due to the Residual.

6.1.5 Two-way analysis of variance

An experiment with two independent variables, or two factors, is called a “two-way design.” For such experiments, the analysis of variance tests for the main effects of each factor on the dependent variable, as well as an interaction effect. A significant interaction effect means that the factors in combination influence, or have an effect on, the dependent variable.

If an experiment has two factors, there are three possibilities for assigning conditions to participants. The assignments may be within-subjects for both factors, between-subjects for both factors, or within-subjects for one factor and between-subjects for the other.

ANOVA Table for Task Completion Time (s)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	11	134.778	12.253				
Device	2	121.028	60.514	5.865	.0091	11.731	.831
Device * Subject	22	226.972	10.317				
Task	1	.889	.889	.076	.7875	.076	.057
Task * Subject	11	128.111	11.646				
Device * Task	2	121.028	60.514	5.435	.0121	10.869	.798
Device * Task * Subject	22	244.972	11.135				

FIGURE 6.15

Two-way analysis of variance for data in [Figure 6.14a](#).

There are three results in Figure 6.15: the device main effect, the task main effect, and the device by task interaction effect.

Example of how to report these results:

The grand mean for task completion time was 15.4 seconds. Device 3 was the fastest at 13.8 seconds, while device 1 was the slowest at 17.0 seconds. The main effect of device on task completion time was statistically significant ($F_{2,22} = 5.865, p < .01$). The task effect was modest, however. Task completion time was 15.6 seconds for task 1. Task 2 was slightly faster at 15.3 seconds; however, the difference was not statistically significant ($F_{1,11} = 0.076, ns$). The results by device and task are shown in Figure x. There was a significant Device $\times$ Task interaction effect ($F_{2,22} = 5.435, p < .05$), which was due solely to the difference between device 1 task 2 and device 3 task 2, as determined by a Scheffé post hoc analysis.

FIGURE 6.16

Results of the two-way analysis of variance in [Figure 6.15](#) as might appear in a research paper.

Statistical tests help assess hypotheses, but they are not the results per se. Give the results in terms of the data, with supporting statistical tests conveyed plainly and without elaboration.

6.1.6 ANOVA tool

You can use Python, R studio or SPSS I think. If anyone wants SPSS, Imk because I accidentally purchased a new one instead of the extension haha so I can give you mine.

6.1.7 Counterbalancing and testing for a group effect

If a factor is assigned within-subjects, learning effects must be offset between conditions, for example by counterbalancing. If counterbalancing worked, there should be no group effect, no significant difference between G1 and G2. (If you want more info on it go back to chapter 5).

- The data can be organized with group as a between-subjects factor and the ANOVA can test for a group effect (a main effect of group on a dependent variable). A non-significant group effect means counterbalancing worked.

6.2 Chi-square test

A common statistical procedure is the **(Pearsons) chi-square** test. The relationships are between categorical, or nominal-scale. The data are commonly summarized in a contingency table. A chi-square test compares the observed values (the counts in the table) against expected values. The expected values are developed under the assumption that there is no difference among the categories in the table.

The chi-square test is a non-parametric test (the categories are nominal-scale attributes and do not have a probability distribution associated with them).

The test statistic is written χ^2 . The underlying assumption is "no difference" with the test, seeking to reject the assumption. The test is simpler, the calculations are easily done using a spreadsheet application.

The chi-square test is a comparison between the observed values and the expected values, with the expected values accounting for the different row and column totals.

Expected value: $\frac{\text{row total} \times \text{column total}}{\text{grand total}}$

Chi-square: $\sqrt{\frac{(\text{observed value} - \text{expected value})}{\text{expected value}}}$

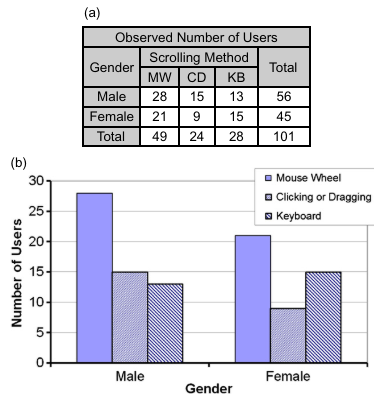


FIGURE 6.22

Contingency table example: (a) Data. (b) Chart.

(a) Expected Number of Users					(b) Chi Squares				
Gender	Scrolling Method			Total	Gender	Scrolling Method			Total
	MW	CD	KB			MW	CD	KB	
Male	27.2	13.3	15.5	56.0	Male	0.025	0.215	0.411	0.651
Female	21.8	10.7	12.5	45.0	Female	0.032	0.268	0.511	0.811
Total	49.0	24.0	28.0	101	Total	0.057	0.483	0.922	1.462

FIGURE 6.23

(a) Expected values and (b) chi-squares for the data in Figure 6.22.

Final chi-square value: the sum of the individual chi-squares in the table.

If the final chi-square exceeds the critical value (in the figure below), the differences in the contingency table are statistically significant.

Significance Threshold (α)	Degrees of Freedom							
	1	2	3	4	5	6	7	8
.1	2.71	4.61	6.25	7.78	9.24	10.65	12.02	13.36
.05	3.84	5.99	7.82	9.49	11.07	12.59	14.07	15.51
.01	6.64	9.21	11.35	13.28	15.09	16.81	18.48	20.09
.001	10.83	13.82	16.27	18.47	20.52	22.46	24.32	26.13

FIGURE 6.24

Chi-square critical values.

The degrees of freedom for the chi-square statistic is $(r - 1)(c - 1)$, where r is the number of rows and c is the number of columns. The alpha level is chosen prior to testing. The critical value for significance is looked up in a table.

If the results show significance and there are more than three categories, a post hoc pairwise comparisons test is required to determine which categories differ from one another.

Relationships studied in HCI with the chi-squared test, are frequently between attributes of people (males versus females, Mac users versus PC users, etc.) and their behaviors (e.g., preferred scrolling method, texting habits, etc.). But, the attributes may also involve systems and the behaviors of systems.

6.3 Non-parametric tests for ordinal data

Non-parametric tests make no assumptions about the probability distribution of the population from which the underlying data are obtained (applicable to a wider range of data).

- Downside to non-parametric tests: loss of information. Most non-parametric tests deal with ordinal data (ranks). If the data tested are in fact interval or ratio, non-parametric tests waste this knowledge by collapsing differences into ranks.
- Non-parametric tests benefit by not being bound by parametric assumptions, but sacrifice the power to use all available information to reject a false null hypothesis.
- The data used for non-parametric tests in HCI are typically obtained through questionnaires (e.g., using a Likert scale), preference ratings, or assessments on a scale.

Design	Conditions	
	2	3 or more
Between-subjects (independent samples)	Mann-Whitney U	Kruskal-Wallis
Within-subjects (correlated samples)	Wilcoxon Signed-Rank	Friedman

FIGURE 6.29

Choosing a non-parametric test based on experiment design and number of conditions.

The four most common non-parametric procedures are the Mann-Whitney U test, the Wilcoxon Signed-Rank test, the Kruskal-Wallis test, and the Friedman test. Each is used in a particular context, depending on the number of test conditions and the experiment design.

Since the Kruskal-Wallis and Friedman tests operate on three or more conditions, a statistically significant outcome is usually followed with a post hoc pairwise comparisons test.

6.3.1 Example 1

Data:

- At least ordinal
- Two conditions
- Between-subjects

→ Mann-Whitney U test. The results are shown in Figure 6.31a.

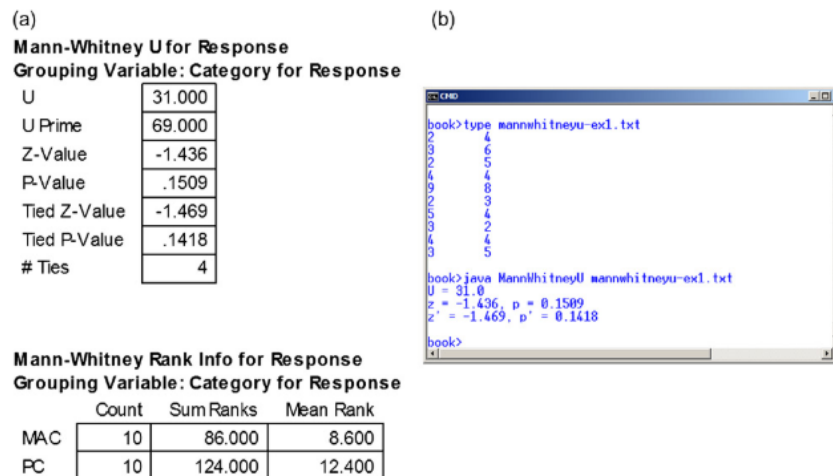


FIGURE 6.31

Mann-Whitney U test for the political leanings of Mac and PC users: (a) StatView. (b) Java utility.

The output also includes a normalized z-score (calculated from U) and p, the probability of obtaining the observed data under the null hypothesis of “no difference”. Two z and p values are provided.

The second set includes a “correction for ties.” In most cases, the difference with the corrected values is minor. For the example, $U = 31.0$ with $p = .1418$ (corrected for ties). Since p is greater than .05, the differences in the assessments have not met the customary threshold for statistical significance.

6.3.2 Example 2

Data:

- Interval-scale
- Two test conditions
- Within-subjects

→ Wilcoxon Signed-Rank test is appropriate.

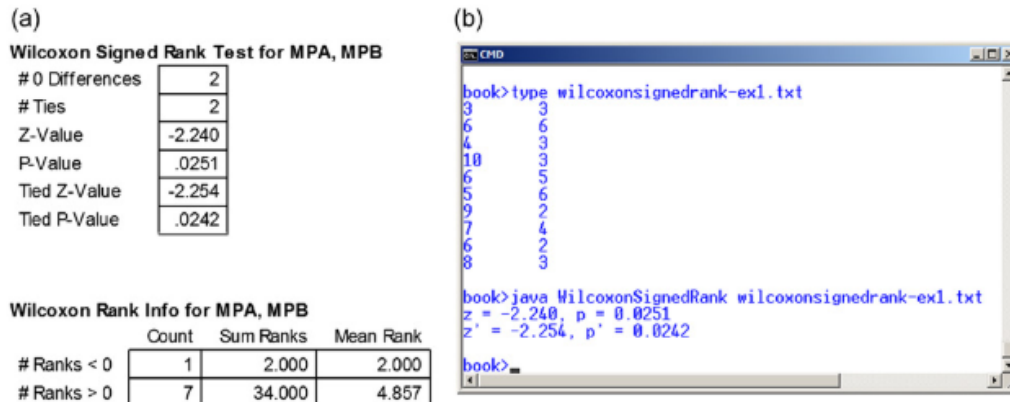


FIGURE 6.33

Wilcoxon Signed-Rank test for the “cool appeal” ratings for two media players: (a) StatView. (b) Java utility.

The test statistic for the Wilcoxon Signed-Rank test is a normalized z score. For the example, $z = -2.254$ with $p = .0242$ (corrected for ties). Since p is less than .05, the customary threshold for statistical significance is exceeded.

6.3.3 Example 3

Data:

- Interval-scale
- Three age groups
- Between-subjects

→ Kruskal-Wallis test is appropriate.

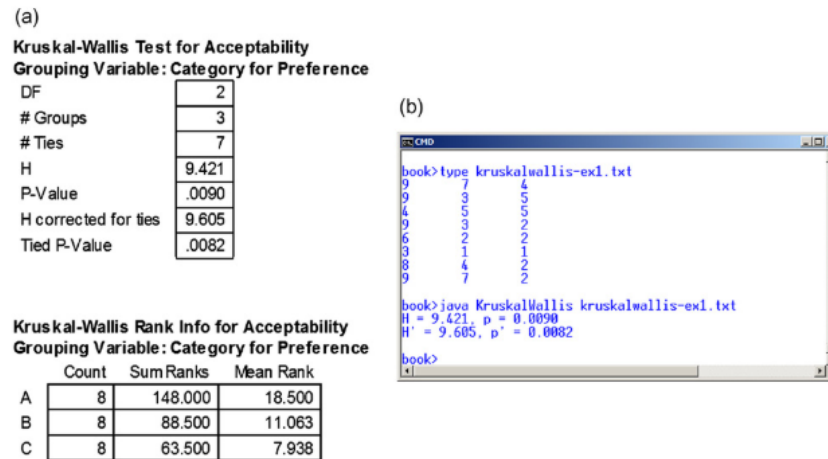


FIGURE 6.35

Kruskal-Wallis test for acceptability of a GPS system by three age groups: (a) StatView.
(b) Java utility.

The test statistic is H , which follows a chi-square distribution with $df = k - 1$ (k is the number of groups). In the example, $H = 9.605$ with $p = .0082$ (corrected for ties). Since p is below the customary threshold for statistical significance (.05), the test shows a significant effect.

There are three age groups, therefore a post hoc pairwise comparisons test is needed to determine which age groups are significantly different from one another.

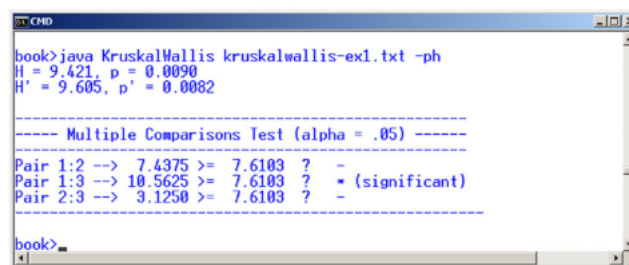


FIGURE 6.36

Post hoc pairwise comparisons for the Kruskal-Wallis test.

6.3.4 Example 4

Data:

- Interval-scale
- Four test conditions

- Within-subjects

→ Friedman test is appropriate

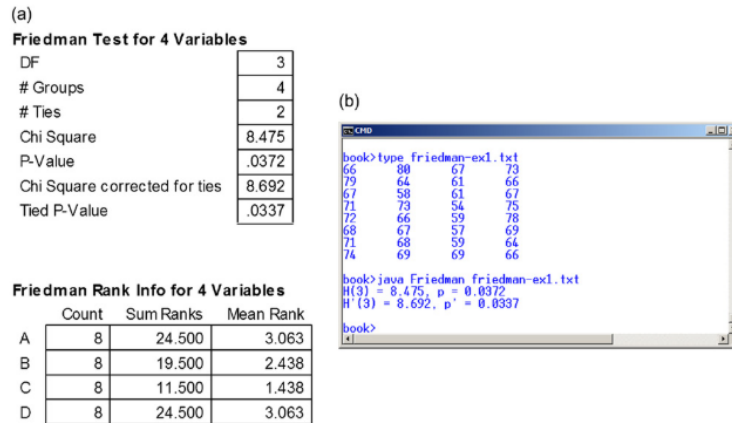


FIGURE 6.38

Friedman test for quality of results for four search engine interfaces: (a) StatView. (b) Java Utility.

The Friedman test statistic is H , which follows the chi-square distribution with $n - 1$ degrees of freedom (n = number of conditions). A post hoc pairwise comparisons test is needed to determine which search engine interfaces are significantly different from one another.

6.3.5 Discussion

In the examples above, the data were summarized using the means (averages) for each condition. This is common and valid in HCI research for data of this sort. The data are interval-scale. Therefore, the mean is the correct statistic for central tendency. However, since the response items in for example likert scales are subject to human interpretation, the data may not be interval-scale in the strictest sense. In the worst case, the data degrade to ordinal and the appropriate statistic for central tendency is the median (middle value) or mode (most common value). In some research papers, questionnaire responses are summarized using the mean, while in other papers the median is used.

Some researchers claim outright that Likert-scale responses are ordinal and that it is simply wrong to report means. The truth is likely somewhere in the middle (i.e., between interval and ordinal).

The non-parametric tests demonstrated above are limited to single-factor analyses. Although there are extensions of non-parametric tests to multi-factor experiments, the value of these tools to HCI research is limited.

6.4 Parametric versus non-parametric tests

Statistical power is about improving the odds of getting it right with the null hypothesis. However, there are other issues to consider:

- Siegel (1957) suggests fitting measurement data type with a statistical test that operates on that type of data.

Parametric tests, such as the analysis of variance, assume the data are sampled from a probability distribution, such as the normal distribution. Provided this assumption is met, parametric tests have more statistical power and are more accurate and more precise than non-parametric tests. However, if the assumptions are not met, there are three possibilities: (1) proceed with the parametric test, (2) transform or clean the data in some manner to correct the violations and then proceed with the parametric test, or (3) use a non-parametric test.

- Transforming data using a log or power function is an effective way to correct the violations in assumptions. But this might cause a distortion in the experimental effect under investigation.
- Proceeding with a non-parametric test sidesteps questions about the probability distribution of the underlying data. However, for ordinal data, non-parametric tests are the logical choice. For ratio data, parametric tests are preferred due to their increased power and by the simple standard that the test-to-data pairing is correct. But do ratio-scale human performance measurements meet the requirements for parametric data? I.e., are measurements such as the time or errors in completing tasks normally distributed?
- If the data are measures on a dependent variable collected in an experiment, a researcher may conclude that the data are not parametric and then proceed with a non-parametric test. But this may be unnecessary or unwise.

In the 1950s and early 1960s, applied statistics in the social sciences (🙌🏻 slay 🙌🏻) experienced a broad migration to non-parametric tests. However, abandoning

parametric tests (mainly the parametric ANOVA) in favor of non-parametric tests was largely unnecessary.

- The shift occurred because researchers asked the wrong question. They asked "Are the assumptions of the parametric ANOVA met?" instead of "What are the consequences of the inevitable violations in the parametric ANOVA?"

There is evidence that the parametric ANOVA is reasonably robust to violations in the underlying assumptions. Even with small sample sizes, the consequences of violations in the assumptions underpinning the ANOVA are usually minor.

Non-parametric tests are well-travelled in the HCI literature. In most cases the tests are used with questionnaire data and the like. That's appropriate since the data are ordinal (or perhaps interval).

Yet there are examples where a non-parametric test was used on ratio data that were obtained experimentally. This use of non-parametric tests merits additional scrutiny.

25 Experiments studied: why were non-parametric tests used?

Sample of the explanations leading to the choice of a non-parametric test:

- "The time data is not normally distributed".
- "The distributions are skewed".
- "[The] Shapiro-Wilk W test for non-normality suggests that the sample is unlikely to be from a normal distribution".
- "[The data] did not meet the requirements for normality".
- "The distribution of task completion times was significantly different from a normal distribution".
- "The data did not fit a normal distribution".
- In none of the papers cited did the researchers articulate concern for the loss of information in using a non-parametric test on ratio data.
- These researchers were evidently concerned with the first of the two questions noted earlier: are the assumptions of the parametric ANOVA met?
- Only two considered: what are the consequences of the violations?

- "Although we tested the data for normality, a non-parametric (distribution-free) test, Friedman's Rank Test for correlated samples, was also run and chi-square scores were computed, in order to confirm the validity of the F-scores obtained through ANOVA".
- "Since the distribution of task completion times was significantly different from a normal distribution, we also used the Mann-Whitney U test for non-parametric statistical significance. However, since the patterns of significant differences ($p < .05$) of the eight tasks were the same for the ANOVA and the Mann-Whitney U, we present the results for the ANOVA".
 - In both of these examples we see a hint of caution (that there is an issue around the use of a non-parametric test versus a parametric test).

Parametric tests might be favored due to the added precision and statistical power. However, examining both avenues, considering the outcome for each, and attune for possible contradictions is also a good approach.

Additional notes week 4

Data

Parametric data:

- Drawn from known (or assumed) distribution, often Normally-distributed
- Interval or, more often, ratio data (measurement of a continuous variable)
- Underlying assumptions are important!

Data can be parametric or nonparametric

- Affects what statistical test needs to be used
- Parametric tests are more powerful

Descriptive Statistics

Measures of Central Tendency

- Reflects the middle or central part of any distribution

- Often referred to as the average
- The most common measures are
 - **Mode** (most frequent value in a data set).
 - Described how most people behave.
 - Pros:
 - Easy to calculate and understand.
 - Can be used with nominal data.
 - Cons:
 - There can be more than one mode.
 - Mode can change dramatically by adding few data points.
 - Independent of all other data in the set.
 - **Median** (Middle value separating the greater and lesser halves of a data set).
 - If scores are even, take the average of the two middle scores.
 - Pros:
 - Relatively unaffected by outliers (very low or high scores) and skewed distributions
 - Can be used with ordinal, interval and ratio data
 - Cons:
 - Does not consider all scores of the data set
 - Can be unstable
 - **Mean** (Sum of values of a data set divided by number of values).
 - Most often used if 'average' is mentioned
 - Pros:
 - Considers every score → most accurate summary of the data

- Resistant to sampling variation: removing one sample changes the mean far less than mode or median
- Cons:
 - Heavily affected by extreme scores and skewed distributions
 - Can only be used with interval and ratio data

Standard Deviation and Variance

- How do we measure variability?
- If $x_1, x_2, \dots, x_n$ are the data in a sample with mean m
 - Mean deviation = difference between mean and scores = $(1/n) * |x_i - m|$
 - Variance $Var(X) = (1/n) * \sum((x_i - m)^2)$
 - Standard deviation $(SD)_s = \sqrt{Var(X)}$
 - We assume a normal distribution.
 - Even when it's not, this often works out!

Confidence Interval (CI)

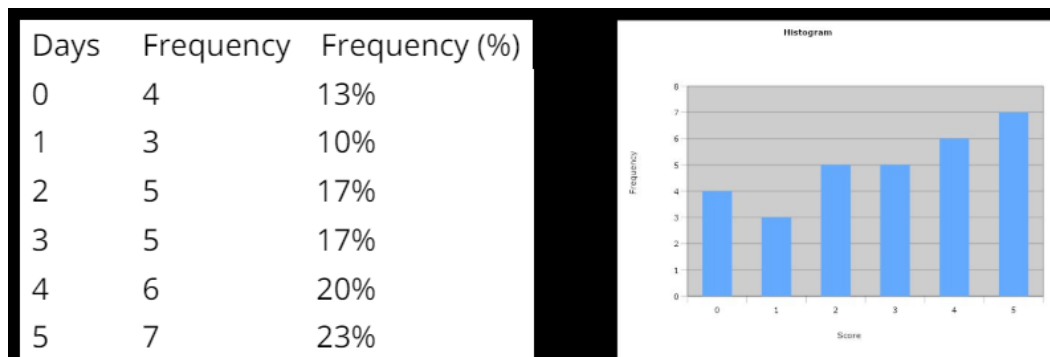
- Confidence interval
 - n% of the cases will lie in this range
 - n is typically 95% or 99%
- We often don't know the true mean or standard deviation
 - Can use a sample
 - Use it's mean and variance to determine a confidence interval on the mean!
 - This generally leads to a good confidence interval on the mean!

Quantile, Quartile and Percentile

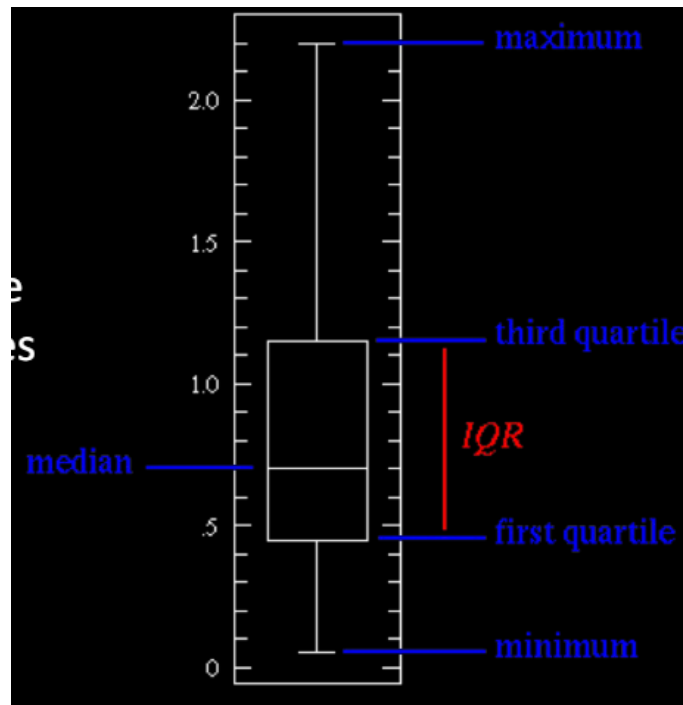
- Quantile

- 'Cut points' that divide a sample of data into groups containing (as far as possible) equal numbers of observations.
- Quartile (Quantile of 4)
 - Values that divide a sample of data into 4 groups containing (as far as possible) equal numbers of observations
- Percentile (Quantile of 100)
 - Values that divide a sample of data into 100 groups containing (as far as possible) equal numbers of observations

Frequency Distributions (Histograms)



Boxplots



Descriptive statistics summarize findings

- Report:
 - Mean and standard deviation in text
 - Bar charts and confidence interval or box plot graphically

Inferential Statistics

Hypothesis: Prediction of the result of an experiment

- Stating how a change in the independent variables will affect the measured dependent variables
- Experiment can support the assumption that the hypothesis is correct
- Usual approach
 - Stating a null-hypothesis (this predicts that the independent variable does not affect the dependent variable)
 - Carrying out the experiment and using statistical measures to reject the null-hypothesis

Statistical significance

- A statistical significant difference between two results exists if the probability that the difference is by chance is below a certain significance level
- Significance level (α)
 - Arbitrary, but...
 - Lower significance level means higher evidence
 - Typical significance levels: $\alpha=0.05$, $\alpha=0.01$
- Significant results ($p < \alpha$)
 - Null hypothesis can be rejected.
 - There is a statistical significant difference between the two results.
- Non-Significant results ($p \geq \alpha$)
 - Null hypothesis cannot be rejected.
 - We cannot conclude anything!
- What is the meaning of α
 - Accepted level of Type I error

Statistical Hypothesis Test: T-Test

- Group of tests for testing level of significance
- Only applicable to interval and ratio data
- Dependent or independent t-test?
 - Dependent: used for repeated measures design
 - Independent: used for independent measures design
- One-tailed or two-tailed?
 - One-tailed test looks for increase or decrease
 - Two-tailed test looks for any change (common choice)

Type I & Type II Errors

p = .028	Type I error (False Positive) Non-existing effect found 2.8%	Correct (True Positive) Existing effect was found 97.2%	true	Effect found in results
	Correct (True Negative) No effect exists, no effect found	Type II error (False Negative) Effect exists but is not found	false	
	false	true		An effect exists

Note on Type II error (beta)

- More difficult to determine
 - Dependent on effect size
 - Can be estimated
 - Indicative of either too little data or rather small effect size
 - We error on the side of caution!

Compare more than 2 conditions

- Do an analysis of variance (ANOVA) & F-Test (omnibus test) first
 - This gives you an indication if there is a chance (and if its valid) to find differences between two result sets
- Do some a priori or post-hoc tests
 - Tests which are aware of the type-1 error cumulation
 - E.g. Tukey test, Scheffé test

Effect sizes

- A significant difference does not tell much about how large that effect of the manipulation is
- Thus, we calculate an effect size

- Tells us which percentage of the difference is due to systematic bias (our manipulation)

Effect sizes: Cohen's d

- Calculated by e.g. Cohen's d
 - Relation between difference of the means and the dispersion (SD)

$$d = \frac{M_1 - M_2}{SD} \quad SD = \sqrt{\left(\frac{SD_1^2 + SD_2^2}{2}\right)}$$

- Widely accepted suggestions are:
 - $d \geq 0.20$ (small effect)
 - $d \geq 0.50$ (medium effect)
 - $d \geq 0.80$ (large effect)

Effect sizes: r (correlation)

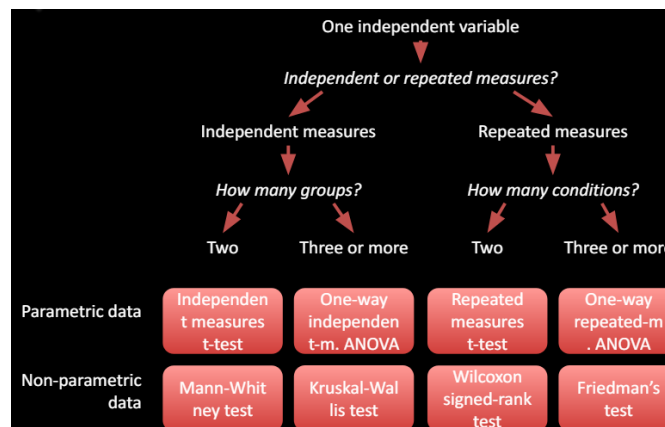
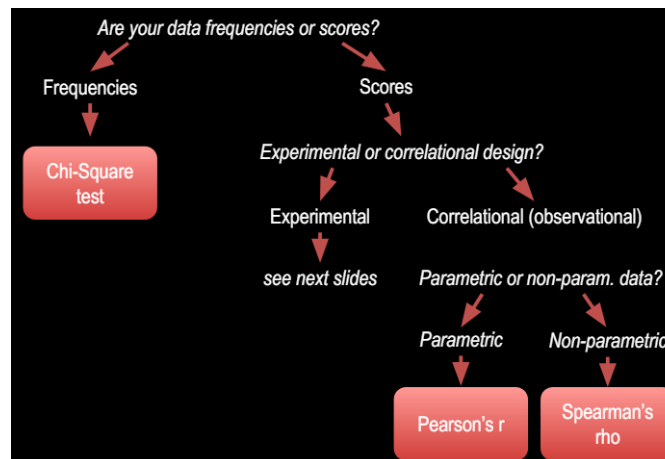
- r = another way to express the effect size
 - Correlation between independent and dependent variables
 - Can be derived from the t-statistic (!= the p value!)
- Can be derived from Cohen's d:

$$r = \frac{d}{\sqrt{d^2 + 4}}$$

- Widely accepted suggestions are:
 - $r \geq 0.10$ (small effect): effect explains 1% of the variance
 - $r \geq 0.30$ (medium effect): effect explains 9% of the variance
 - $r \geq 0.50$ (large effect): effect explains 25% of the variance

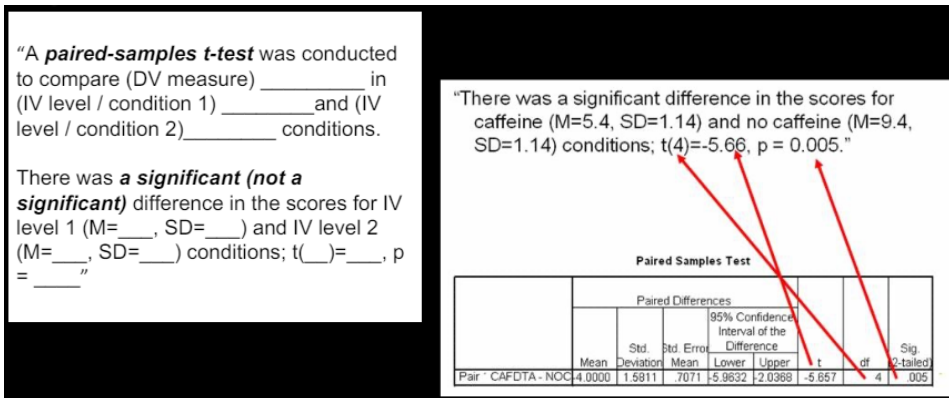
Choosing the Right Test

Data Setup	Parametric test	Non-Parametric test
1 Variable 2 Categories Between Subjects	independent t-test	Mann-Whitney U test
1 Variable 2 Categories Within-Subjects	paired t-test	Wilcoxon Signed Rank Test
1 Variable >2 Categories Between Subjects	One-way ANOVA	Kruskal Wallis Test
1 Variable >2 Categories Within Subjects	repeated measures ANOVA	Friedman test Mood's median test
1 Variable (Correlation)	Pearson's r	Spearman's ρ (rho)



Additional notes lecture week 5

How do I report my test results? (APA format)



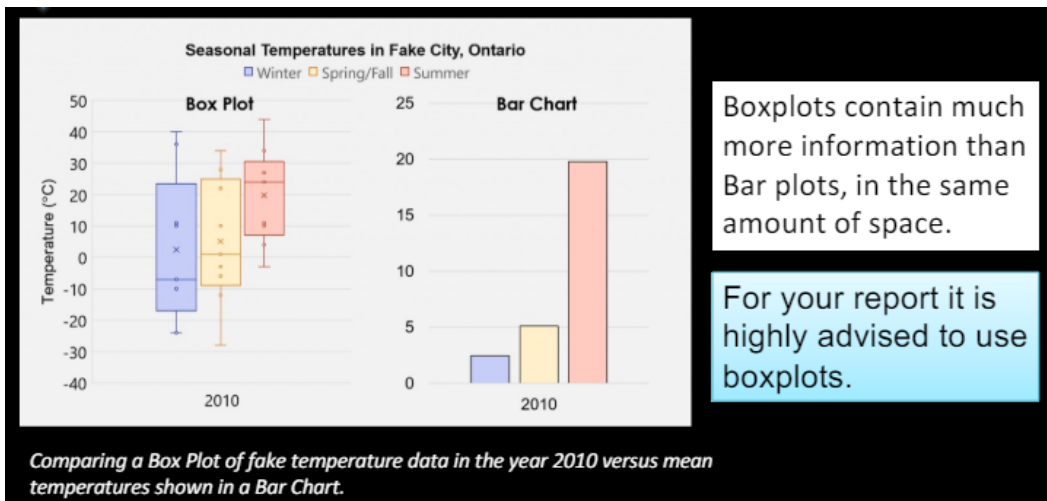
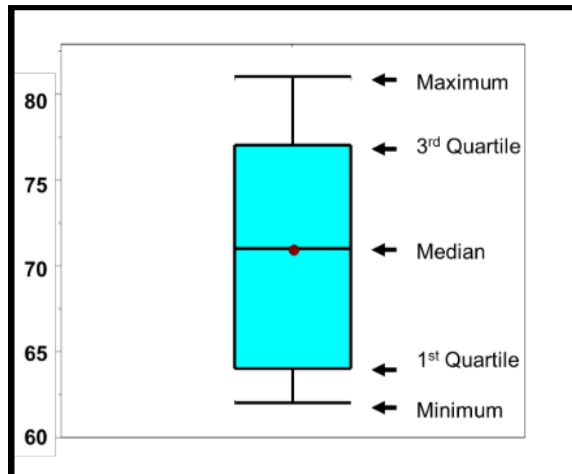
Data Visualization

Tables vs. Figures

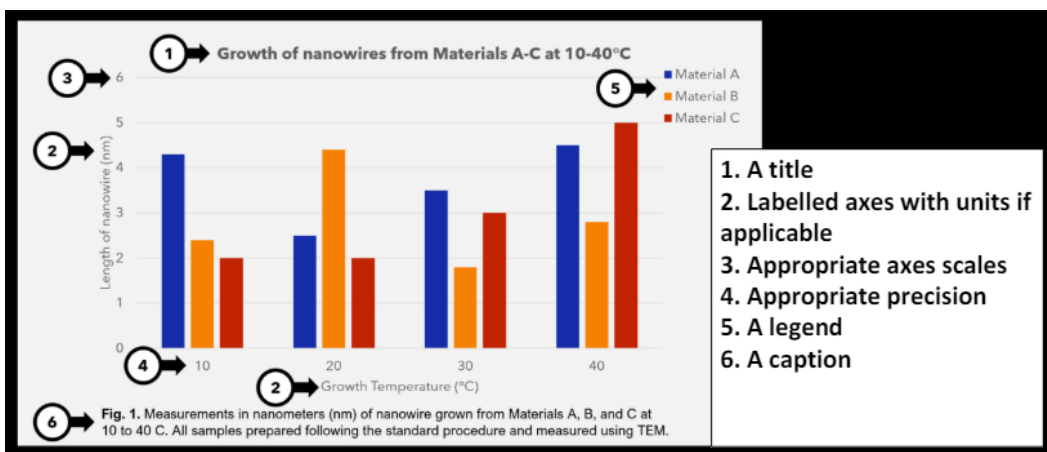
- Tables are perfect for grouping the data and presenting detailed information and precise values for each variable.
- Figures are suitable for showing the trends, patterns and relationships in the data. They are visually more engaging and also more effective in conveying high-level information.

Bar plots and boxplots

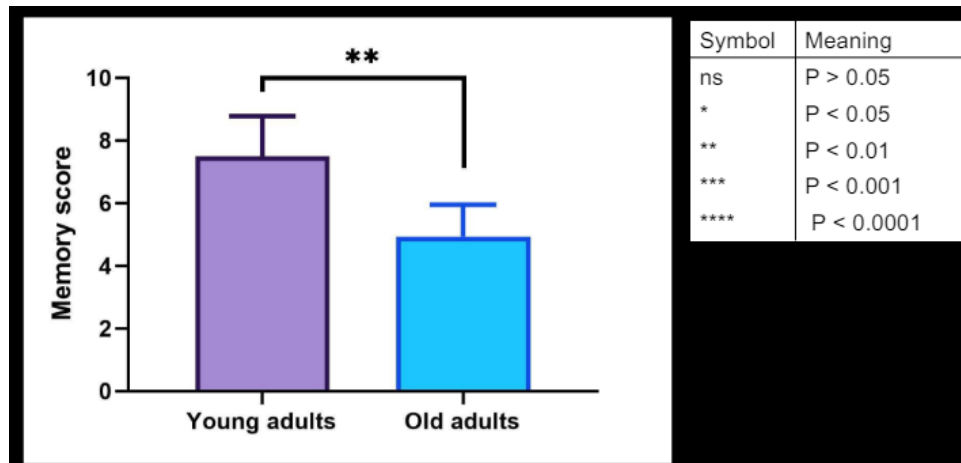
- Bar plots are commonly used for comparing values between different categories or groups.
- Boxplots are used for visualizing the distribution and spread of a dataset.
 - When visualizing normally distributed data, you can use bar plots to show mean and SD.
 - When visualizing non-normal data, you should use boxplots.



Essential parts of a figure



Showing statistical significance on plots



Chapter 7 - Modeling Interaction

Model: a simplification of reality, reductions or simplifications of more complex phenomena. They allow us to explore the phenomena, think about them, make changes, and so on, without actually constructing or doing it.

Models using "loose verbal analogy and metaphor" describe phenomena → **descriptive models**.

Models using "closed-form mathematical equations" predict phenomena → **predictive models**.

7.1 Descriptive models

7.1.1 Delineating a problem space

A descriptive model can be as simple as a dividing up a problem space. You can make it a partitioned domain.

Example:

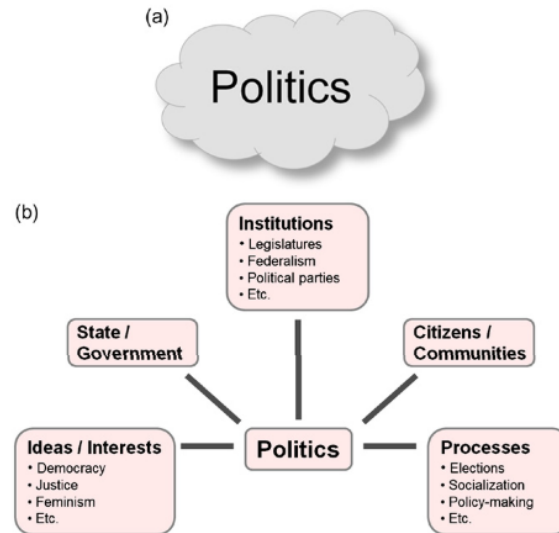


FIGURE 7.1

Politics: (a) The “big fuzzy cloud” model. (b) A descriptive model delineating the problem space.

Power of descriptive models: with every tweak, refinement, and improvement we get a better model and, more importantly, we get a better understanding of the problem.

7.1.2 Quadrant model of groupware

An important area of research within HCI is computer supported collaborative work (CSCW). Computer applications to support collaboration are known as groupware.

Example:

	Same Time	Different Times
Same Place	Copy boards PC Projectors Facilitation Services Group Decision Room Polling Systems	Shared Files Shift Work Kiosks Team Rooms Group Displays
Different Places	Conference Calls Graphics and Audio Screen Sharing Video Teleconferencing Spontaneous Meetings	Group Writing Computer Conferencing Conversational Structuring Forms Management Group Voice Mail

FIGURE 7.2

Quadrant model of groupware.

(Adapted from Johansen, 1991, Figure 1)

New technologies have been developed now, and can probably be added to the model.

One use of a descriptive model is to identify common elements for interactions located in the same area of the problem space.

7.1.3 Key-action model (KAM)

Model about the operation and organization of keys on a keyboard.

Graphical representation of the model:

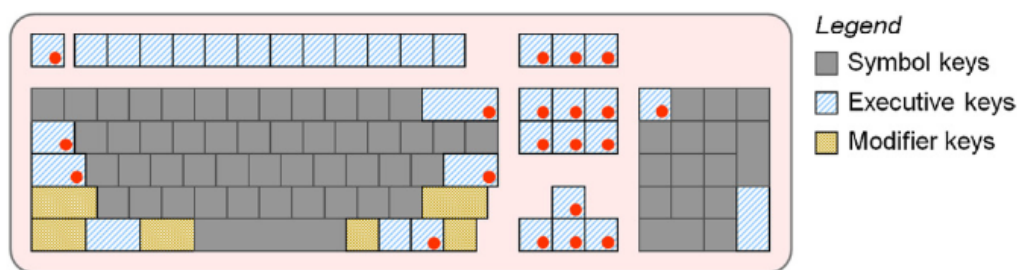


FIGURE 7.3

The key-action model (KAM) illustrated. Keys marked with (red) dots are executive keys that are not mirrored on both sides.

Keys without a (right-side) replica, are identified by dots.

There is no greater measure of the merit of a model than its ability to tease out critical arguments on the potential, the capabilities, and the limitations in an interaction domain.

The key-action model only captures one aspect of keyboards, namely the actions associated with each key. Another way to think about keyboards is in the ambiguity of key presses.

7.1.4 Model of bimanual control

Example: "a simple model, based on a non-quantitative physical approach, which aims at describing the logic of the division of labor that appears to govern the variety of human bimanual asymmetrical actions". This is the essence of a descriptive model.

Hand	Role and Action
Non-preferred	• Leads the preferred hand • Sets the spatial frame of reference for the preferred hand • Performs coarse movements
Preferred	• Follows the non-preferred hand • Works within established frame of reference set by the non-preferred hand • Performs fine movements

FIGURE 7.4

Guiard's model of bimanual control.

Task	Characteristics
Scrolling	• Precedes/overlaps other tasks • Sets the frame of reference • Minimal precision needed (coarse)
Selecting, editing, reading, drawing, etc.	• Follows/overlaps scrolling • Works within frame of reference set by scrolling • Demands precision (fine)

FIGURE 7.6

Relationship between scrolling and common GUI tasks.

Scrolling is in conflict with a basic goal of good user interfaces: unobtrusiveness and transparency. Users should not be required to divert their attention from the primary task to acquire and manipulate user interface widgets.

- Scrolling is an appropriate task for the non-dominant hand and that this is revealed through Guiard's model of bimanual control.

7.1.5 Three-state model for graphical input

Buxton's three-state model of graphical input is described as "a vocabulary to recognize and explore the relationship between pointing devices and the interaction techniques they afford".

The model seems simple and obvious, and we might question its ability to add insight to the existing body of pointing device research. Yet the model can be extended to capture additional aspects of pointing device interaction.

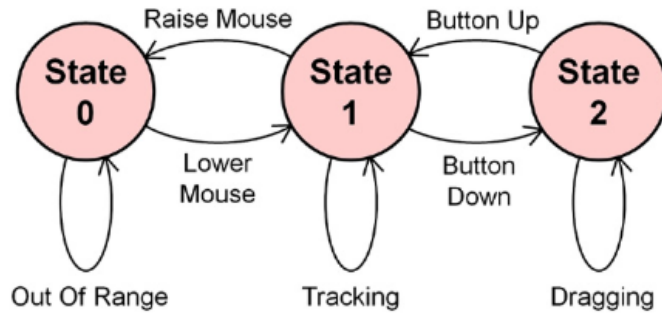


FIGURE 7.9

Buxton's three-state model for graphical input.

Mackenzie and Oniszczak used the three-state model to help characterize an interaction technique that didn't exist at the time Buxton's model was introduced.

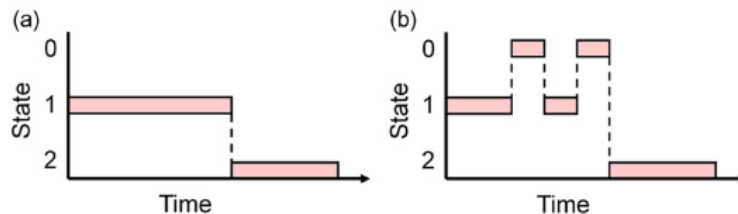


FIGURE 7.10

State transitions for dragging tasks: (a) Mouse. (b) Lift-and-tap touchpad. Dragging begins upon entering State 2.

7.2 Predictive models

Predictive model: an equation. The equation predicts the outcome of a variable based on the value of one or more other variables (predictors). The outcome variable is a dependent variable. The only requirement is that the variable use continuous, or ratio-scale, data.

The predictor variable can not be called an independent variable.

- In experimental research, independent variable has a special meaning: it is a circumstance or characteristic that is manipulated. The vast majority of independent variables are nominal-scale attributes.
 - This poses a problem: a nominal-scale variable cannot serve as a predictor in a prediction equation. Of course, independent variables in experimental research can also be ratio-scale attributes. Such variables

can serve as predictors in prediction equations. A predictor variable can also be a ratio-scale attribute of users.

Predictive models allow a problem space to be explored.

7.2.1 linear regression model

Linear relationship between an independent variable (x , a predictor variable) and a dependent variable (y , a criterion variable or human response)

$$y = mx + b$$

(Where m is the slope of the relationship and b is the y intercept.)

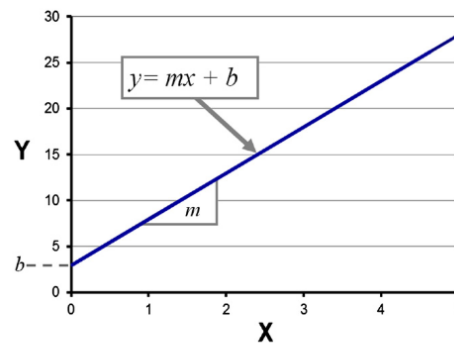


FIGURE 7.11

Linear relationship between an independent variable (x) and a dependent variable (y).

To build this equation, we first need a set of x - y sample points. Sample points will scatter about. If the model is good, the points will be reasonably close to a straight line.

Finding the best-fitting straight line involves a process known in statistics as linear regression. The objective is to find coefficients m and b for the line that minimizes the squared distances (least squares) of the points from the line. If the data are good and the model is correct, a good prediction equation should emerge. There is also a built-in assumption that the relationship is linear, which is not necessarily the case.

Example:

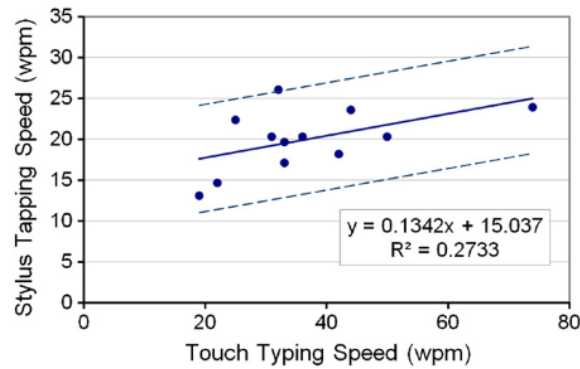


FIGURE 7.13

Scatter plot from Figure 7.12b embellished with linear regression line, prediction equation, squared coefficient of correlation (R^2) and dashed lines showing the 95% confidence interval.

R^2 : the amount of variation in the data that is explained by the model. It is commonly articulated as a percent.

One of the benefits of a predictive model is the potential to predict the outcome on a value of the predictor never actually visited.

There is clearly some uncertainty surrounding this prediction. SE (standard error) establishes confidence intervals around a prediction. Dashed lines show the 95% confidence interval window (95 percent of the times, the sample will fall within this area).

- Usually linear regression models are reported giving the equation and R^2 .
- Confidence intervals and SE are generally not given (there are exceptions).

7.2.2 Fitts' law

If an interaction involves a rapid-aimed movement, such as moving a finger or cursor to a target and selecting the target, there's a good chance someone has experimented with the interaction and used Fitts' law to model it.

Three uses in HCI:

1. To learn if a device or interaction technique conforms to the model by building the prediction equation and examining the correlation for "goodness of fit,"
2. To use the prediction equation in analyzing design alternatives,
3. To use Fitts' index of performance as a dependent variable in a comparative evaluation.

7.2.2.1 background

Fitts argued that the amplitude of an aimed movement was analogous to the information in an electronic signal and that the spatial accuracy of the move was analogous to electronic noise. He proposed that the human motor system is like a communications channel, where movements are like the transmission of signals. Fitts' analogy expresses the information capacity C (in bits/s) of a communications channel of bandwidth B (in s^{-1} or Hz) as:

$$C = B \log_2 \left(\frac{S}{N} + 1 \right)$$

- C : information capacity
- B : bandwidth
- S : signal power
- N : noise power

Experiment:

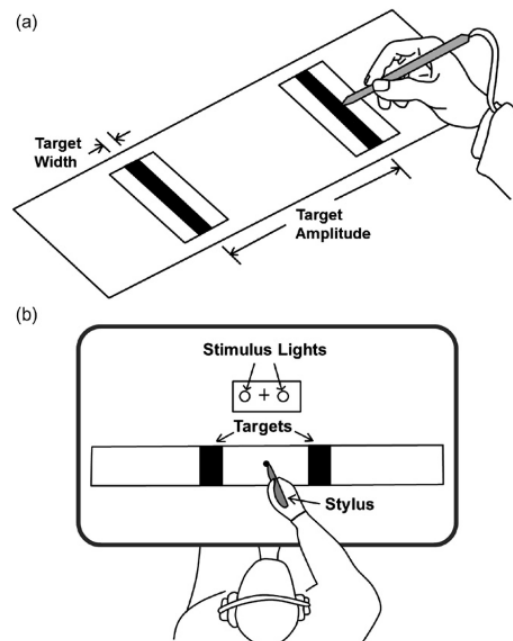


FIGURE 7.14

Experimental paradigm for Fitts' law: (a) Serial task. (b) Discrete task.

Subjects selected one of two targets in response to a stimulus light. The relationship Fitts examined is that between the amplitude of the movement task, the time taken, and the width, or tolerance, of the region within which the move

terminates. Fitts summarized his findings as follows: "The results are sufficiently uniform to indicate that the hypothesized relation between speed, amplitude, and tolerance may be a general one". Fitts proposed to quantify a movement task's difficulty (ID, the index of difficulty) using information theory by the metric "bits.":

$$ID = \log_2 \left(\frac{2A}{W} \right)$$

- A: amplitude
- W: width
- Doubling the distance to a target has the same effect as halving its size.

The amplitude (A) and width (W) in Equation 5 are analogous to Shannon's signal (S) and noise (N).

The following expression for ID was introduced later to improve the information-theoretic analogy:

$$ID = \log_2 \left(\frac{A}{W} + 1 \right)$$

Fitts' index of performance, now called throughput (TP, in bits/s), is calculated by dividing ID (bits) by the mean movement time, MT (seconds), computed over a block of trials:

$$TP = \frac{ID_e}{MT}$$

The subscript e in ID_e reflects a small but important adjustment, which Fitts endorsed in his 1964 paper. An adjustment for accuracy involves first computing the effective target width as

$$W_e = 4.133 \times SD_x$$

- SDx: standard deviation in a participant's selection coordinates.
- We: includes the spatial variability, or accuracy, in responses. In essence, it captures what a participant actually did, rather than what he or she was asked to do.

This adjustment necessitates a similar adjustment to ID, yielding an "effective index of difficulty":

$$ID_e = \log_2 \left(\frac{A}{W_e} + 1 \right)$$

If using Fitts' law as a predictive model is the goal, then the movement time (MT) to complete a task is predicted using a simple linear equation:

$$MT = a + b \times ID$$

The slope and intercept coefficients in the prediction equation are determined through empirical tests, typically using linear regression.

7.2.2.2 Example (idk where 7.2.2.1 went tbh)

You can read this in the book if you want.

7.2.3 Choice reaction time

Given n stimuli, associated one-for-one with n responses, the time to react (RT) to the onset of a stimulus and choose the correct response is given by:

$$RT = a + b \log_2(n + 1)$$

where a and b are empirically determined constants. Reasonable values for the constants are $a \approx 200\text{ms}$ and $b \approx 150\text{ms/bit}$.

Typical setup for a choice reaction time experiment:

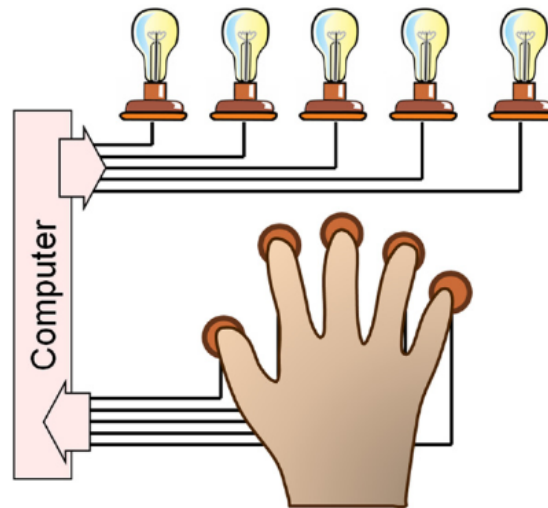


FIGURE 7.18

Paradigm for choice reaction time experiments.

As with Fitts' law, the log term is analogous to the information content of the task and has units of "bits." If some of the choices are more probable than others, the information content of the task is reduced. This in turn reduces the choice reaction time.

For a set of alternatives with different probabilities, the information (H) is

$$H = \sum_i p_i \log_2 \left(\frac{1}{p_i} + 1 \right)$$

- p_i : the probability of occurrence of the i th item in the set.

The Hick-Hyman law has failed to gain the same momentum in HCI as Fitts' law.

- The model alone is of limited use. In most situations there are additional behaviors coincident with choice reaction, such as movement or visual search and these complicate applying the model.

7.2.4 The keystroke-level mode

The keystroke-level model (KLM) was developed specifically for analyzing human performance in interactive computing systems. The model predicts expert error-free task completion times using the following elements:

- Task (or series of sub-tasks)
- Method used
- Command language of the system
- Motor skill parameters of the user
- Response time parameters of the system

The model is useful in situations where the sequence of interactions in performing a task is known. The KLM is not useful if the details of the interactions are not known. A related model, GOMS (Goals, Operators, Methods, Selection rules), operates at a higher level and attempts, among other things, to predict the method the user will adopt.

The total predicted time is the sum of the subtask times. The model works with four motor-control operators (K = keystroking, P = pointing, H = homing, D = drawing), one mental operator (M), and one system response operator (R):

$$t_{\text{EXECUTE}} = t_K + t_P + t_H + t_D + t_M + t_R$$

Some operations are omitted or repeated, depending on the task. The pointing operator, t_P , is a constant based on Fitts' law.

7.2.4.1 Original KLM experiment

To validate the KLM, Card, Moran, and Newell conducted an experiment using 14 tasks performed using various methods. Task T1, for example, was "Replace one 5-letter word with another (one line from previous task)." It was performed on three different systems, each with a unique command set.

The task required four mental operations (M) and 15 keystroking operations (K):

$$t_{\text{EXECUTE}} = 4 \times t_M + 15 \times t_K$$

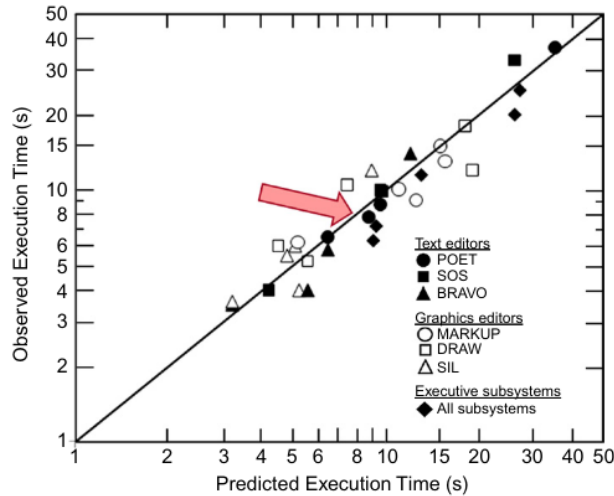


FIGURE 7.21

Observed versus predicted execution times for tasks in the KLM validation experiment. The arrow shows task T1 with POET.

This provides a general validation of the model.

A predictive model is built on parameters. If parameters become variables and slide back and forth, what is the effect on the model's outcome? How sensitive are the predictions to changes in the parameters? This was explored in a series of arguments and demonstrations.

An example for an editing task:

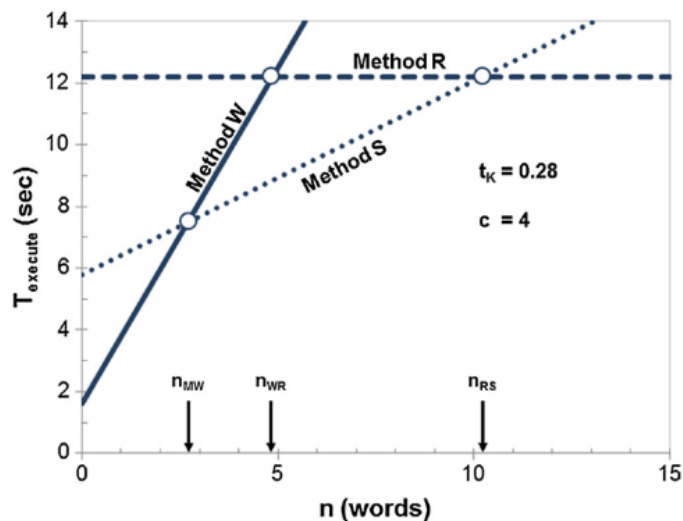


FIGURE 7.22

Parametric analysis showing the sensitivity of predictions for three methods based on changes in n —the location of an edit relative to the current location.

If the designer is choosing among candidate methods, and it is known that certain values of n are common, then the parametric analysis in Figure 7.22 provides precisely the information needed.

A model is a malleable tool.

7.2.4.2 Modern applications of the KLM

- Early experiments with the KLM related primarily to text editing and document management.
 - e.g. “delete a line of text,” “add a label to a box,” or “transfer a file to another computer, renaming it”
 - The available task methods are quite different now.

KLM model is old, changes should be made:

- With the ubiquity of the mouse in today’s GUIs, the pointing operator could benefit from the additional fidelity afforded with a Fitts’ law prediction. A reasonable substitute for P is

$$t_p = 0.159 + 0.204 \times \log_2 \left(\frac{A}{W} + 1 \right)$$

The prediction above includes the terminating mouse-button click to select the target, unlike the original P operator.

Much work extends the KLM by adding or refining operators to suit the application.

Use of the KLM entails breaking a task into subtasks and building an overall prediction using the equation, perhaps updated with new or refined operators. If the model is being extended or applied in a novel setting as part of a research initiative, then validation is required.

- This is typically done in a formal or semi-formal experiment, where the task is performed with users while task completion times are observed and measured. The predicted and observed times are compared.
 - Deviations between predictions and observations signal a problem, but not necessarily a problem in the KLM.

- May indicate a problem in the task coding.

Predictive models like the KLM are also tools for thinking, for generating insight.

- It is difficult to get inside a user's mind to know what he or she is contemplating, let alone attribute a time cost to mental processes in a model to use the KLM to model interactions in novel HCI contexts.
- Arguably, the mental operator (M) in the KLM is its Achilles heel. It is a single operator, pegged at 1.35 s, and is used for all mental processes. The human mind is simply too rich to be distilled so uniformly.
 - For this reason, researchers often do not model or predict execution time, but instead, model or predict only the keystrokes or other primitive actions.

7.2.4.3 The KLM and (Predictive) text entry

A classic HCI example of keystroke-level modeling, minus the hidden costs, is text entry, particularly mobile text entry. The mobile phone keyboard, being ambiguous, bears a special challenge for text entry. The relevant statistic for this is KSPC (keystrokes per character), defined as the number of keystrokes required, on average, for each character of text entered using a given input method in a given language.

- A mobile phone has KSPC ≈ 2.023 for multi-tap and KSPC ≈ 1.007 for predictive text entry (T9). If word completion or word prediction is added, KSPC < 1 .

Some pictures about research with KLM and text entry:

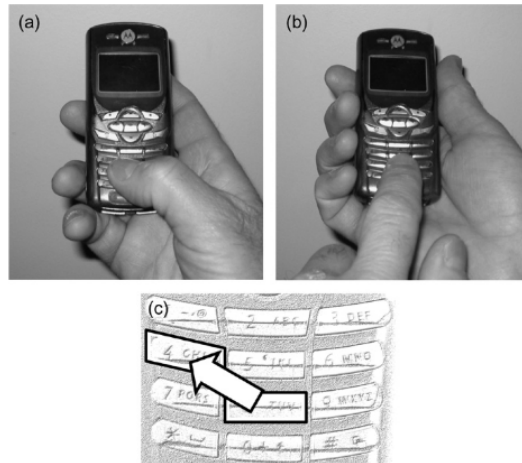


FIGURE 7.27

Interaction with a mobile phone keypad: (a) Thumb. (b) Index finger. (c) Fitts' law movement for "h" preceded by "t."

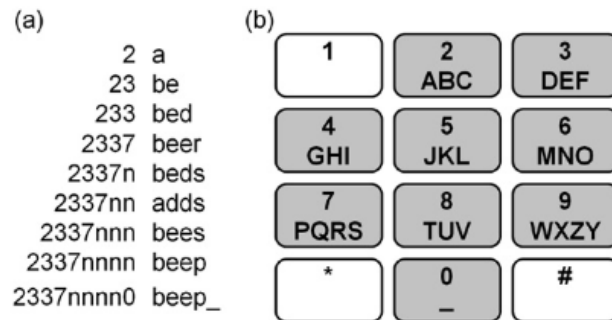


FIGURE 7.28

(a) Key sequence (*left*) and display progress (*right*) for input of *beep* using predictive text entry. Note: *n* is the *next* key (e.g., *). (b) Mobile phone keypad (for reference).



FIGURE 7.29

Key sequences and mental operators for the input of "beep" (see Figure 7.28).

Rank	Word	Keystrokes	Higher Ranking Colliding Word (rank)
47	if	43n0	he (15)
51	no	66n0	on (13)
63	then	8436n0	them (57)
72	me	63n0	of (2)
78	these	84373n0	there (35)
105	go	46n0	in (6)
118	us	87n0	up (56)
159	home	4663n0	good (115)
227	night	64448n0	might (141)
298	war	927n0	was (10)

FIGURE 7.30

Top ten ambiguous words requiring one press of next.

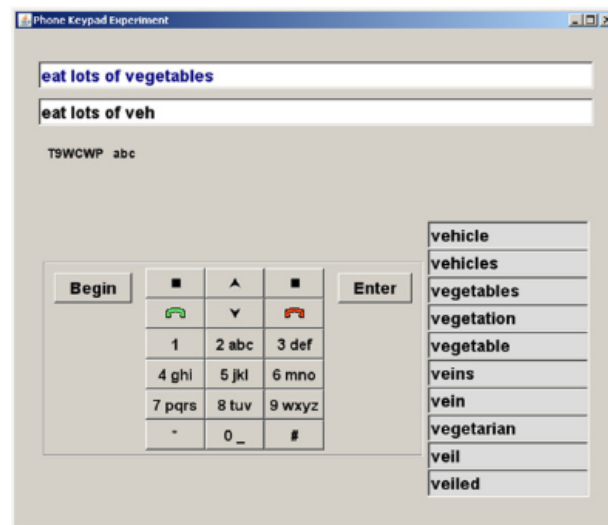


FIGURE 7.31

Input of *vegetables* using a GUI simulation of a phone keypad with word completion. After three key presses (834), *vegetables* appears as the third entry in the candidate list.

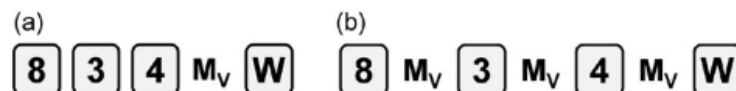


FIGURE 7.32

Key sequences and mental operators for input of *vegetables* (see Figure 7.31): (a) Visual check after third keystroke. (b) Visual check after each keystroke.

7.2.4.4 Updating the KLM's mental operator (M)

MP and MV were suggested as mental operations where the cognitive process involves physical matching or visual searching. But what are the execution times?

The KLM's mental operator was not supported by any theory.

- The value of M was estimated from the experimental data "by removing the predicted time for all physical operations from the observed execution time". Then tM was estimated by a least-squares fit of the time removed versus the estimated number of mental operations. The result was $tM = 1.35$ s (SD = 1.1s).
- M_P , physical matching, was motivated by Card et al.'s model human processor, which includes a range of human behaviors called "simple decisions." These behaviors are not included in the KLM. Each behavior involves a stimulus that is connected to a response through one or more cycles of cognitive processing. The behaviors are given in Figure 7.33, along with a proposed mnemonic.

Proposed Mnemonic	Task	Execution Time (ms)	
		Card et al.	Figure 2-28 & Figure 2-30
M_S	Simple Reaction	240 [105 – 470]	277 [± 44]
M_P	Physical Matching	310 [130 – 640]	510 [± 59]
M_N	Name Matching	380 [155 – 810]	485 [± 52]
M_L	Class Matching	450 [180 – 980]	566 [± 96]
M_C	Choice Reaction	$200 + 150 \log_2(N + 1)$	
M_V	Visual Search		$498 + 41 N$

FIGURE 7.33

Mental operators and proposed mnemonics for simple decision tasks, along with the nominal value and the range of execution times (ms).

The execution times include the time for motor response.

Card et al.'s values were obtained by reviewing a large body of literature on human sensory-motor responses.

All values are well below 1.35 seconds, the value for the KLM's original mental operator (M).

The visual search operator (M_V) cannot be expressed as a nominal value, or even a range of values. The time depends on the number of choices the user must scan.

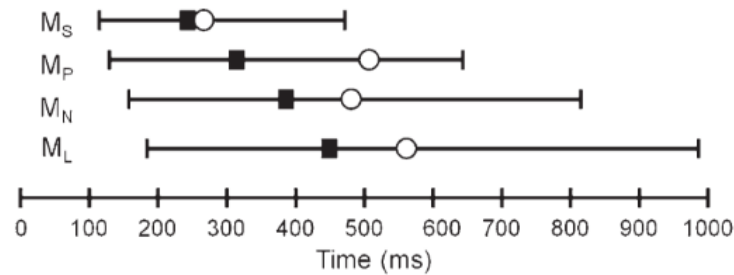


FIGURE 7.34

Nominal value and range of times (ms) for M_S, M_P, M_N, and M_L operators. Times are from Card et al. (1983, 65–76). Circles are means from experiment results in Figure 2.28 and summarized in Figure 7.33.

7.2.5 Skill acquisition

There is a relationship between skill and practice. If the skilled behavior is adequately represented by a ratio-scale variable, then the transition from novice to expert is well suited to predictive modeling.

- Predicted or dependent variable: performance or skill.
- Independent variable: amount of practice.
- The relationship between skill and practice is non-linear. In the beginning, a small amount of practice yields substantial improvement. After a lot of practice, the same "small amount" produces only a slight improvement. Mathematical expression of the relationship:

$$y = b \times x^a$$

- x: regressor or independent variable (amount of practice)
- y: dependent variable (performance)
- a and b are constants that determine the shape of the relationship
- Represented in this form, the relationship between skill and practice is often called the power law of learning (or practice).
- Written in terms of time:

$$T_n = T_1 \times n^a$$

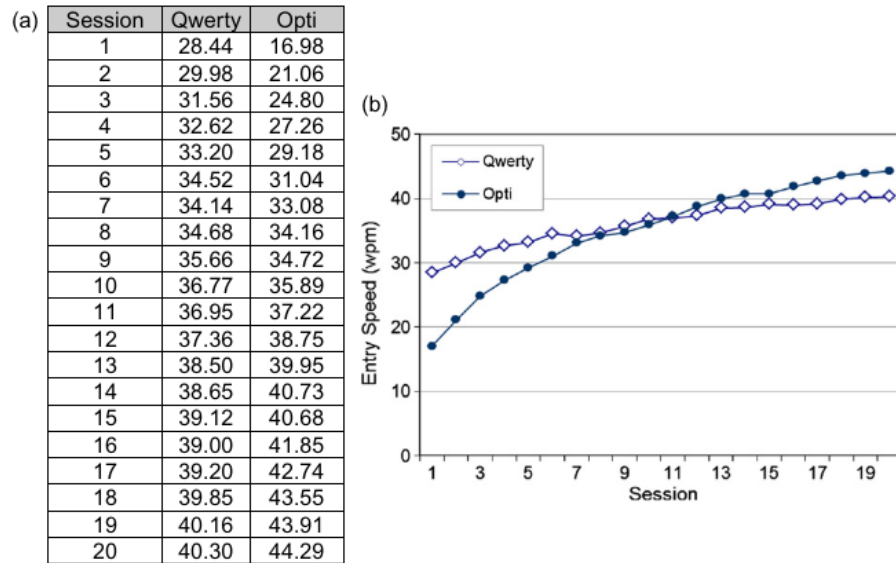


FIGURE 7.36

Experiment demonstrating skill acquisition: (a) Data showing entry speed (wpm) by session. (b) Chart.

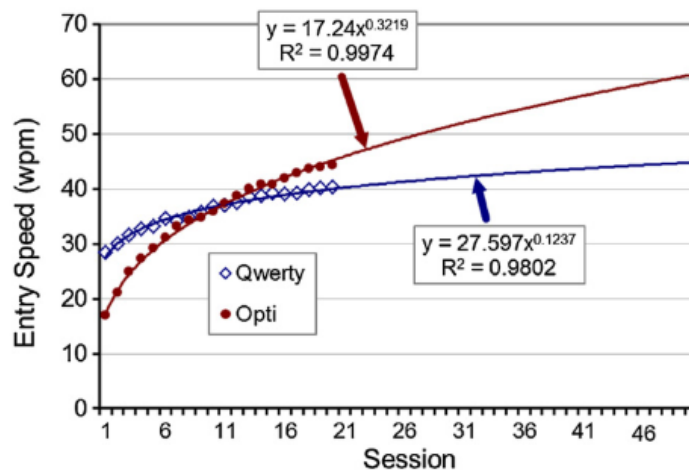


FIGURE 7.37

Power law of learning for Opti and Qwerty showing extrapolation to session 50.

7.2.5.1 Skill acquisition research in HCI

7.2.6 More than one predictor

The equations above predict the outcome on a dependent variable from a single predictor variable (independent variable). It is also possible to build a prediction model where the outcome is predicted from multiple predictors. In statistics the technique is called multiple regression and takes the form

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + \dots$$

Example:

- The dependent variable is y , the time to reach a performance criterion in playing the game.
- The two independent variables (predictors) are x_1 , a player's age, and x_2 , a player's daily computer use in hours.

Another example:

Model for MT (ms) ^a	Fit ^b	Variance Explained
$MT = 435 + 190 ID_e$	$r = .560$	31.30%
$MT = 894 + 46 LAG$	$r = .630$	39.80%
$MT = -42 + 246 ID_e + 3.4 LAG$	$R = .948$	89.80%
$MT = 230 + (169 + 1.03 LAG) ID_e$	$R = .987$	93.50%
^a LAG in ms, ID_e in bits		
^b $n = 48$, $p < .0001$ for all models		

FIGURE 7.41

Models for movement time (MT) based on index of difficulty (ID_e) and system lag (LAG).

At lag = 0, the fourth model reduces to

$$MT = 230 + 169 \times ID_e$$

which is consistent with other Fitts' law models for the mouse.

HCI researchers with a social science perspective tend to use multiple regression with a slightly different motivation. Social scientists are more interested in observing and explaining human behavior rather than in measuring and predicting human performance.

Additional notes week 6

Descriptive Models

- Descriptive models

- provide a basis for discussing, understanding, reflecting, and reasoning about certain facts and interactions
- provide a conceptual framework that simplifies a system
- are used to inspect an idea or a system and make statements about their probable characteristics
- used to reflect on a certain subject
- can reveal flaws in the design and style of interaction

Predictive Models

Predictive (causal) models

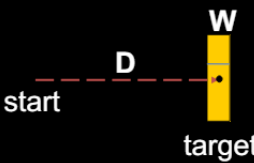
- Equation that predicts outcome (probability of an outcome) based on the value of one or more predictor variables
 - Useful to explain findings in past user experiments (model fitting)
 - Useful to predict findings in future experiments (model validation)
 - Equation can be simple (e.g. linear regression) or very complex

Fitts' law

- Developed by Paul Fitts in 1954
- Robust model of human psychomotor behavior
- Predicts movement time for rapid, aimed pointing tasks
- Fitts' discovery "was a major factor leading to the mouse's commercial introduction by Xerox" [Stuart Card]

Fitts' Law – Equation (inspired on Shannon–Hartley theorem)

- The time to acquire a target is a function of the distance to and size of the target and depends on the particular pointing system

$$MT = a + b \log_2 \left(1 + \frac{D}{W} \right)$$


- MT: movement time
- a and b: constants dependent on the pointing system
- D: distance to the target area
- W: width of the target

Fitts' Law – Index of Difficulty

$$MT = a + b \log_2 \left(1 + \frac{D}{W} \right)$$

- Index of Difficulty, $ID = \log_2 \left(1 + \frac{D}{W} \right)$
- $MT = a + b \cdot ID$
- ID describes the difficulty of the task independent of the device / method
- Units
 - Constant a measured in seconds
 - Constant b measured in seconds / bit
 - Index of Difficulty, ID measured in bits

Target 1

$ID_1 = ID_2$

Target 2

(Simple) Linear Regression

- How to measure a and b for a new pointing device / menu / etc.?

$$MT = a + b \log_2 \left(1 + \frac{D}{W} \right) \quad ID = \log_2 \left(1 + \frac{D}{W} \right)$$

- Setup an experiment with varying D and W and measure MT
- Fit a line through the measured points: a = intercept, b = slope

a: intercept

b: slope

$y = 82.71x + 716.63$
 $R^2 = 0.169$

Image from: Heiko Drewes. 2010. Only one Fitts' law formula please!. In CHI '10 Extended Abstracts on Human Factors in Computing Systems (CHI EA '10). ACM, New York, NY, USA, 2813-2822.

Fitts' Law variants

- Extension to 2D
 - "Status Quo": use horizontal width
 - "Sum Model": $W = \text{width} + \text{height}$
 - "Area Model": $W = \text{width} * \text{height}$
 - "Smaller Of": $W = (\text{width}, \text{height})$
 - "W' Model": width in movement direction

Implications for HCI

- Bigger buttons
- Proportional to amount of use?!
 - See principle (and golden rule) of consistency!
- Use current location of the cursor
 - distance is close to zero
- Use edges and corners
 - edges of the screen have infinite height or width, respectively
 - corners have infinite height and width

Hick's Law (Hick-Hyman Law)

- The time needed to make a selection is proportional to the log number of alternatives given
- $T = a + b * H$
- H is the information-theoretic entropy of a decision
- $H = \log_2(n + 1)$.
n alternatives of equal probability
- Common practical values: $a=200\text{ms}$, $b=150 \text{ ms/bit}$
- Hick's law does not apply if it requires linear search. It applies if the user can search by sub-division

The GOMS Model

- G: goals
 - (Verbal) description of what a user wants to accomplish
 - Various levels of complexity possible
- O: operators
 - Possible actions in the system
 - Various levels of abstraction possible (sub-goals / ... / keystrokes)
- M: methods
 - Sequences of operators that achieve a goal
- S: selection rules
 - Rules that define when a user employs which method (among alternatives)
- User tasks are split into goals which are achieved by solving sub-goals in a divide-and-conquer fashion

Keystroke-Level Model

- Simplified version of GOMS
 - Only operators on keystroke-level
 - No sub-goals
 - No methods
 - No selection rules
- KLM predicts how much time it takes to execute a task
- Execution of a task is decomposed into primitive operators:
 - Physical motor operators
 - Mental operator
 - System response operator

Models: Levels of Detail

- Different levels of detail for the steps of a task performed by a user
- Abstract: correct wrong spelling
- Concrete: mark-word
delete-word
type-word
- Keystroke-Level: hold-shift
n-cursor-right
recall-word
del-key
n-letter-key

KLM Operators

- Each operator is assigned a duration (amount of time a user would take to perform it):

K	Key press
B/BB	Mouse button press / click
P	Mouse pointing
D	Mouse drawing
H	Move hand between mouse and keyboard
M	Thinking (mental act)
R	Response time from the system

- Each operator is assigned a duration

Operator	Description	Associated Time
K	Keystroke, typing one letter, number, etc. or function key like 'CTRL', 'SHIFT'	Expert typist (90 wpm): 0.12 sec Average skilled typist (55 wpm): 0.20 sec Average non-secretarial typist (40 wpm): 0.28 sec Worst typist (unfamiliar with keyboard): 1.2 sec
H	'Homing', moving the hand between mouse and keyboard	0.4 sec
B / BB	Pressing / clicking a mouse button	0.1 sec / 2*0.1 sec
P	Pointing with the mouse to a target	0.8 to 1.5 sec with an average of 1.1 sec Can also use Fitts' Law
D(n_D, l_D)	Drawing n _D straight line segments of length l _D	$0.9 * n_D + 0.16 * l_D$
M	Subsumed time for mental acts; sometimes used as 'look-at'	1.35 sec (1.2 sec according to [Olson and Olson 1995])
R(t) or W(t)	System response (or 'work') time, time during which the user cannot act	Dependent on the system, to be determined on a system-by-system basis

Predicting the Task Execution Time

Execution Time

- OP: set of operators
- n_{op}: number of occurrences of operator op

Example task on Keystroke-Level:

- hold-shift
- n · cursor-right
- recall-word
- del-key
- n · letter-key

Sequence:

K (Key)
n · K
M (Mental Thinking)
K
n · K

$$T_{execute} = \sum_{op \in OP} n_{op} \cdot op$$

Operator Time Values: K = 0.28 sec. and M = 1.35 sec
 $2n \cdot K + 2 \cdot K + M = 2n \cdot 0.28 + 1.91 \text{ sec}$

→ time it takes to replace a n=7 letter word: T = 5.83 sec

GOMS vs. KLM

(CMN-)GOMS	KLM
• Pseudo-code (no formal syntax) • Very flexible • Goals and subgoals • Methods are informal programs • Selection rules ⇒ tree structure: use different branches for different scenarios • Time consuming to create	• Simplified version of GOMS • Only operators on keystroke-level ⇒ focus on very low level tasks • No multiple goals • No methods • No selection rules ⇒ strictly sequential • Quick and easy
Problem with GOMS in general	
• Only for well defined routine cognitive tasks • Assumes statistical experts • Does not consider slips or errors, fatigue, social surroundings, ...	

See slides for KLM examples

Using KLM

- KLM can help evaluate UI designs, interaction methods and trade-offs
- If common tasks take too long or consist of too many statements, shortcuts can be provided
- Predictions are mostly remarkable accurate: +/- 20%
- Sometimes already the comparison of the number of occurrences of the different operators for different designs reveal the difference

Weaknesses of GOMS et al.

- Difficult to target specific users
- No real users
- Difficult to model novel interactions
- Various variable parameters
- Users like to have impact

Strengths of GOMS et al.

- Gives reasons
- Helps in decision making

- Identifies bottlenecks
- Provides illustrative figures
- Combines various views
- Treats feasibility and cognitive load
- Less cost in money and time

Chapter 8 - Writing and Publishing a Research Paper

8.1 Conference papers, journal papers

The main venues for publishing HCI research papers are conference proceedings or journals. Journals are considered the top tier for research publications. Two primary reasons:

1. They undergo peer review to ensure the research is novel, correct, and carried out according to accepted standards in the field. The reviewers are experts in the subject matter and are enlisted by the journal's editor or by a member of the editorial board.
2. Journals are archived in major libraries around the world; thus, research published in a journal is readily available to other researchers.

HCI conferences bring together researchers and practitioners. The practitioners are the engineers and designers of products, and they are there to learn about the latest research in HCI.

The timeline from submission to publication in a journal is long, typically one to two years, sometimes more. For conferences the timeline is short, typically eight months, sometimes less.

- Publishing in a conference proceedings is the quickest way to get one's work "out there."
- The archival advantage for journals no longer exists. Today, the term archive extends to online databases maintained by organizations such as the ACM or

IEEE.

- The review process is also different. For some HCI conferences, the peer review process is as rigorous as it is with many journals. So the prestige of publishing in some HCI conferences, such as the ACM's annual SIGCHI conference, equals that of publishing in some journals.

Prior to publication, a research paper is a manuscript. In most cases, the venue for submitting is decided before writing begins.

Journals generally have relaxed requirements for the format of submissions.

- This is reasonable since a journal submission, if accepted, undergoes a round or two of revision before publication. Revisions are introduced to improve the manuscript based on the referees' suggestions. Acceptance is often conditional on final approval by the journal editor or referees. Publication in a journal also includes professional copyediting by the journal staff.

Conferences, on the other hand, have strict requirements on the format of submissions, including page length.

1. The timeline is short. The entire process is deadline-driven, with dates that are set for the initial submission, reviewer feedback, notification (accept or reject), and final submission.
2. The responsibility lies with the author(s) to provide the final camera-ready copy that is published.

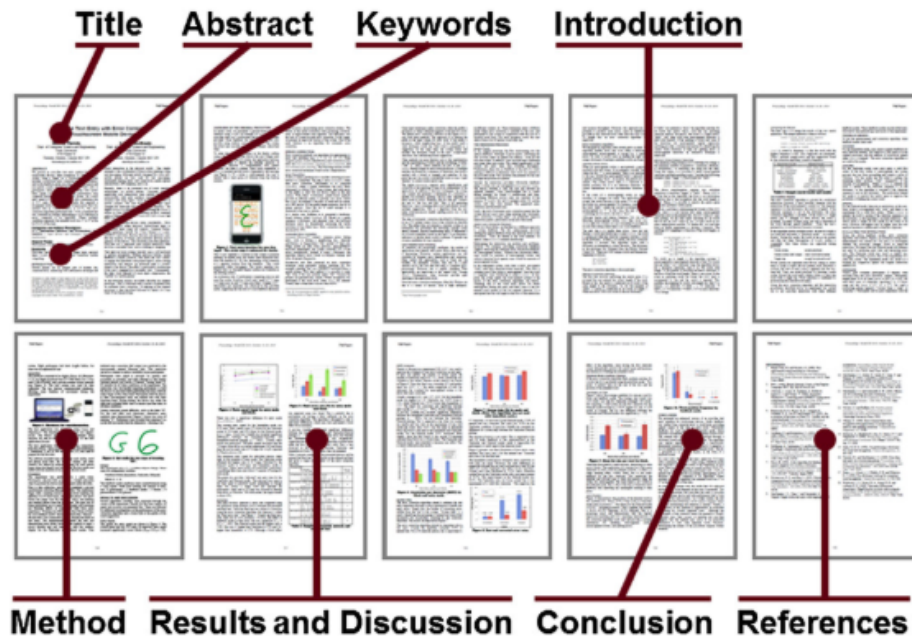


FIGURE 8.1

Parts of a research paper. The backdrop paper is a 10-page conference contribution by Tinwala and MacKenzie (2010).

8.2 Parts of a research paper

8.2.1 Titles

Any title is fine provided it is concise, identifies the subject area, and narrows the scope of the work.

- An overly broad title fails to indicate if the paper is relevant to a sub-topic within a subject area and is misleading.

The title is followed by the names of, affiliations with, and contact information for the authors. This information is provided and positioned according to the submission requirements of the conference or journal.

8.2.2 Abstract

- Written last
- Typically a size limit imposed, such as 150 words, so the abstract must be concise.
- A single paragraph and should not include any citations.

- Delivers a succinct summary of the story within. After reading the abstract, the reader will decide if the rest of the paper is relevant and worth reading.
 - The abstract should be the best-written section of the paper.
- Should tell the reader what was done and what was found.
- The abstract is not an introduction to the subject matter of the paper.
- Common flaw: failure to give specific results.

8.2.3 Keywords

- Used for database indexing and searching.
- Identify the subject matter and the scope of the work.
- Sometimes it is required to add a CCS.

Categories and Subject Descriptors

H.5.2 [Information Interfaces and Presentation]: *User Interfaces*
 – *input devices and strategies (e.g., mouse, touchscreen)*

General Terms

Performance, Design, Experimentation, Human Factors

FIGURE 8.2

Example of the ACM Computing Classification System.

8.2.4 Introduction

- Opening section of a research paper.
- Gives the context for the research.
 - E.g. relevance
- A user interface problem or challenge is identified and the reader is alerted, early on, to the solution that is developed in the rest of the paper.
- Common to give an overview of the contents of the entire paper.

In the following section, we briefly describe our original prototype. This is followed with a review of related work on automatic error correction. A redesign of the original prototype is then described followed by details of a user study to test the prototype.

FIGURE 8.3

Provide an overview of the entire paper early in the Introduction.

- Desirable to state the contribution of the work.
 - Usually it is sufficient to note that the idea improves on existing practice, or something similar.
- Usually a literature review is expected. This is typically organized in a separate section, with an appropriate heading (e.g., Related Work).
 - Discusses earlier published work related to the subject matter of the research.
 - Points relevant to the current research are presented.
- Usually finishes with a statement indicating the need to test the idea in an empirical evaluation.

8.2.5 Method

- Tells the reader how the experiment was designed and carried out.
- Should be written in an entirely straightforward fashion (like a recipe).
- Delivers reproducibility.
- Predictability is important. It allows a reader to scour papers and quickly find key points and results.
 - Needs the following subsections (and in the following order):
 - Participants
 - Apparatus
 - Procedure
 - Design

8.2.5.1 Participants

- States the number of participants and how they were selected.
- Relevant demographic information is also given, such as age, gender, and related experience. Other details might be useful.
- The information on participants identifies the population of people used and, therefore, the population to which the results apply.
- Usually short, just a couple of sentences. However, if a property of the user is an independent variable, then the participants section is often more detailed.

8.2.5.2 Apparatus

- Describes the hardware and software.
- Again, reproducibility.
- If the experiment used a custom interface, the development language (e.g., Java) and other relevant details should be given. It is particularly useful to provide screen snaps or photos of the interface.

8.2.5.3 Procedure

- Tells exactly what happened with each participant.
 - State the instructions given, and indicate if demonstration or practice was used, etc.
 - Typically, participants are tested over multiple conditions and perhaps multiple trials for each condition. Give the details.
 - Define tasks properly.

8.2.5.4 Design

- Summarizes the experiment in terms of the independent variables (factors and levels), the dependent variables (measures and units), or other relevant details.
- It is common to begin with a statement such as, "The experiment was a 3×2 within-subjects design."
- If counterbalancing was used, the way the conditions were administered to participants should be stated.

- A good way to conclude the design section is to indicate the total number of trials administered in the experiment.

The feedback mode conditions were counterbalanced using a Latin square. Aside from training, the amount of entry was $12 \text{ participants} \times 3 \text{ feedback modes} \times 3 \text{ blocks} \times 4 \text{ phrases/block} = 432 \text{ phrases}$.

FIGURE 8.4

A calculation reveals the total number of trials in the experiment.

8.2.6 Results and discussion

It is common in HCI to combine the results with discussion. (Sometimes a discussion section follows the results section.)

- If there were outliers or problems in the data collection, state this up front and give details on any filtering of the data that took place before analysis.
- Before presenting the results it is common to describe the statistical approach and tests used in the data analysis.
- It is common to use subsections that organize the results by dependent measures, beginning with the most important dependent variable.
- For each dependent variable, begin with a broad observation, such as the overall mean, and then move to finer details such as the means for each test condition. The difference in the means between test conditions is the effect size.
 - Example:

The mean task completion time for method A was 2.7 seconds. Method B was 9.1% slower with a mean task completion time of 3.0 seconds.

FIGURE 8.5

Reporting results in absolute and relative terms.

- It is important to explain the results through discussion.

- Testing for a real difference typically involves doing an analysis of variance on the data. The results of this test are given in a succinct statement indicating the outcome.
 - In conveying the ANOVA results, bear in mind that the statistical tests are not the results per se. The results lie in the observations and measurements.
 - The result is the difference in outcome between the test conditions.
- Giving too many results or too much data is an indication that you can't distinguish what is important from what is unimportant.
 - If there is no insight to be gleaned from a result, then don't include it.
- Another way to make results interesting for the reader is through visuals.
- Should compare the results with those in other research papers on a similar topic, citing prior work as appropriate.
- Summarizes information obtained through questionnaires or interviews given at the end of testing.

8.2.7 Conclusion

- Summarizes what you did, restates the important findings, and restates the contribution.
- Common to identify topics for future work, although developing new ideas is to be avoided in the concluding section.
- Often followed by an acknowledgment thanking funding sources as well as people who assisted in some way with the research.

8.2.8 References

Contains the full bibliographic details on papers cited earlier in the paper.

8.3 Preparing the manuscript

8.3.1 Formatting

Formatting is about: the punctuation, spelling, capitalization, italics, quotations, abbreviations, numbers, variables, and so on. These properties of a manuscript

are important for clarity and flow.

The Publication Manual of the American Psychological Association (APA) is recommended.

- Style website: www.apastyle.org

To facilitate formatting, conferences provide template files, typically in both Microsoft Word and LaTeX formats.

8.3.2 Citations and references

Research submissions have formatting requirements for citations and reference lists.

- HCI conferences generally deviate from the APA guidelines.

8.3.2.1 Reference list

1. Aula, A., Khan, R. M., and Guan, Z., How does search behavior change as search becomes more difficult? <i>Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems - CHI 2010</i> , (New York: ACM, 2010), 35-44.	Conference paper
2. Brajnik, G., Yesilada, Y., and Harper, S., The expertise effect of web accessibility evaluation methods, <i>Human-Computer Interaction</i> , 26, 2011, 246-283.	Journal paper
3. Brown, T., <i>Change by design: How design thinking transforms organizations and inspires innovation</i> . New York: HarperCollins, 2009.	Book
4. Buxton, W., There's more to interaction than meets the eye: Some issues in manual input, in <i>User centered system design: New perspectives on human-computer interaction</i> , (D. A. Norman and S. W. Draper, Eds.). Hillsdale, NJ: Erlbaum, 1986, 319-337.	Chapter in book
5. ESA, <i>Electronic Software Association, Industry facts</i> , http://www.theesa.com/facts/ , (accessed February 4, 2012).	Internet document

FIGURE 8.6

Formatting examples for common types of publications in a reference list in a conference proceedings.

- References are numbered.
- References are ordered alphabetically by first author's surname.
- For each author, the surname comes first, followed a comma, then the initials for the given names. Include a space between the initials if there is more than

one (e.g., "Smith, B. A." not "Smith, B.A.").

- For the title of the publication, only capitalize the first word, the first word in a secondary title (e.g., after a colon), and proper nouns.
- Always include the year. Substitute "in press" for accepted but not-yet-published papers.
- Always include page numbers (except for complete books or web pages).
- The name of the publication is set in italics with all keywords capitalized (e.g., *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems—CHI 2011*).
- For journal publications, include the volume number in italics.
- If space permits, use the full name for conferences and journals. If space is tight, use abbreviated names for conferences and journals (e.g., *Proc CHI '99*). Do not mix full and abbreviated names; use one style or the other. If using abbreviated names, be consistent.
- Give the location and name of the publisher for conference papers and books (e.g., "New York: ACM").
- Use align left (ragged right) for the reference list. (Note: The rest of the manuscript is justified.)
- Only include works that are cited in papers.
- Study and imitate!
- Be consistent.

8.3.2.2 Citations

Connect current research with prior research or other sources. Most commonly for conference submissions, citations appear as numbers in brackets.

- Do not treat citations as nouns, except if it is inside parentheses.
- Include all the authors' names for the first appearance of the citation and use "et al." for subsequent appearances. Of course, all the authors' names appear in the reference list.

Tool: www.endnote.com

8.3.3 Visual aids

One of the most common results to present is the effect of an independent variable on a dependent variable.

- A bar chart is usually the most appropriate format, since most independent variables are nominal-scale attributes.
- If the variable shown along the x-axis is continuous, then a line chart is appropriate.

The methodology in particular can benefit from a judicious and targeted use of visuals.

- A powerful way to augment the procedure section of a research paper is through a photograph of a participant performing the experiment task.

Summaries of Likert-scale questionnaire items may be presented as shown in Figure 8.9.

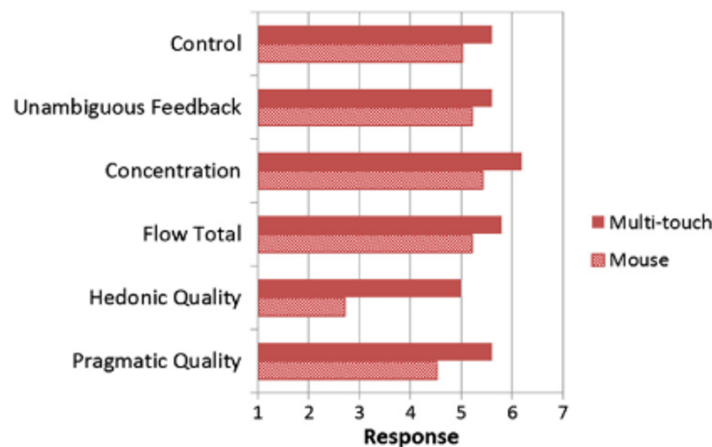


FIGURE 8.9

A chart summarizing participant responses from a questionnaire on target acquisition using multi-touch versus a mouse.

8.3.4 Writing for clarity

The goal in writing a research paper is communication. Effective communication demands clarity.

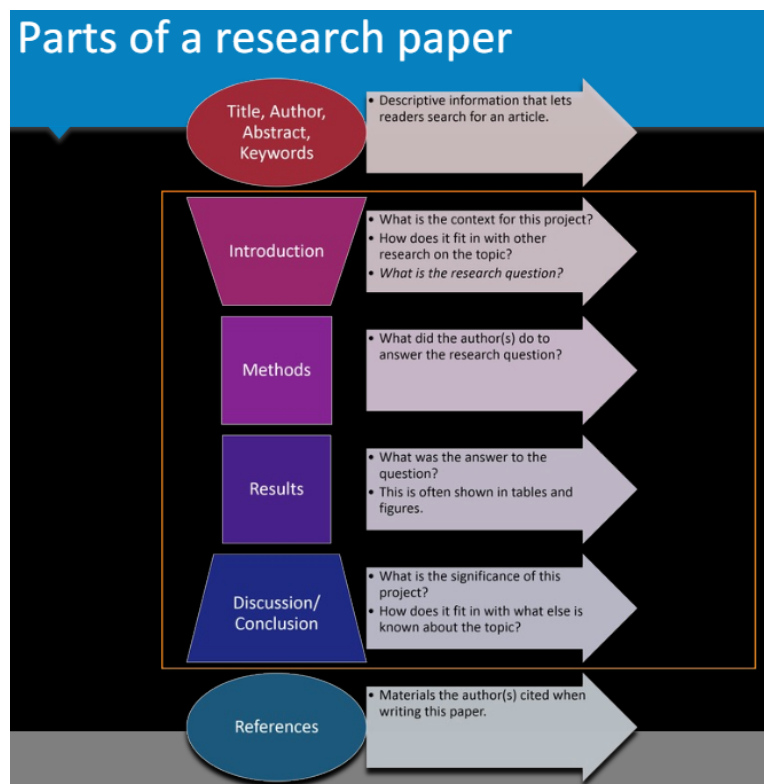
- Clarity comes with economy, saying what needs to be said and little else.
- Write in a straightforward style.

- Avoid long or complex sentence structures.
- One technique: get rid of clutter.

Additional notes week 7

Conference vs. Journal

- Both journals and conferences serve as platforms for dissemination of research findings.
- Depending on the domain, they differ significantly in their publication process (manuscript format, review timeline, etc.), evaluation criteria, and impact.
- Journal papers undergo a rigorous peer-review process, emphasizing in-depth analysis, while conference papers focus on novel developments and preliminary findings.
- In the HCI domain, some conferences such as SIGCHI are so prestigious and competitive that having a manuscript accepted there is deemed as important as publishing in a journal.



Parts of a research paper

- Each part of the manuscript serves a purpose:
 - Title: tells it all in one sentence.
 - Not too short and not too long.
 - Not too broad and not too specific.
 - People often try to make it catchy!
 - Abstract: A one paragraph summary of the whole study that is often written last.
 - The abstract should tell the reader what was done and what was found.
 - Follows the same structure; intro, methods, results and implications.
 - For journal publications, the abstract is the most important element when it comes to the editor's first impression and decision.
 - Keywords: 3 to 6 most important terms that describe the theme of the research paper.
 - Is used for database indexing and searching
 - Example: Child-Robot Interaction, Mathematics learning, robot tutors, Motivational feedback, Facial expressions
 - Introduction: The opening of the paper where you give the problem statement, motivation of research and the impact.
 - You can choose to present your literature review in the introduction or as a separate section e.g., "background" or "related works".
 - The intro should be balanced in terms of breath and width of the reviewed topics.
 - Method: documents what you actually did in your study. Did you design a new interface? Did you test the interface in a user study? Whichever it is you should document all the steps you took so that your study is replicable.

- The methods section usually has multiple subheadings where you break down the information for the reader.
- How you structure the information is up to you for as long as the information flow is understandable for the reader.
- An HCI user study report often has the following subsections:
 - Participants
 - Experimental design
 - Materials and apparatus
 - Experimental procedure
 - Measurements
 - Analysis
 - etc
- Remember! The language of the Methods (and Results section) is most often in past tense!
- Results: Where you present the findings of your study and data analysis (e.g., data from questionnaires or interviews + statistical analysis)
 - It is common practice to use subsections that organize the results by dependent measures (Order them consistently with the order you defined them in the Methods section).
 - You must use visual aids (figures, tables, etc.) to make the results more engaging and understandable for the reader.
 - The Results and Discussion sections are at the same level as Method (So, You are not allowed to introduce results in the Methods section).
 - Also, you are not allowed to introduce new methods information in the results section.
- Discussion: Tries to interpret the results reported in the Results section and make sense of them (What caused the differences in the measurements across experimental conditions? If no differences were observed, what could have been the reason? What were the limitations of

the study, interface design or measurements? What could be improved in the future?)

- The discussion section should link the findings of the study with previous literature supported by references. (was the interaction design better/more effective/etc.? Was the RQ answered? Was the gap in the lit. filled?)
- Some time Discussion and Conclusion are presented together in some papers. It's generally advised to have a separate conclusion section at the end.
- You are not allowed to introduce new results in the Discussion
- Conclusion: summarizes what the study did and restates the important findings.
 - Conclusion is like an abstract with more focus placed on the contribution of the paper to the field and suggestions for future.
 - Like abstract it shouldn't exceed one paragraph.
- References: a list of papers that were used and cited in the paper.
 - Only papers cited in the paper are included in the reference list.
 - If a paper is included in the references, it should be cited within text.
 - References and citations should be formatted as stipulated by the conference or journal. e.g., References can be numbered or ordered alphabetically according to the first author's surname.

YESSSSSSSS I am finally done with this summary, and so are you! Well done!!!!!! 🙌

Good luck with the exams!